

Adverse Selection Among Early Adopters: A Dynamic Free-Rider Problem in New Markets

Rory McGee*

July 2, 2026

Abstract

I study “selection markets,” where consumers differ in willingness-to-pay and the costs they impose on sellers. I give primitive conditions for stronger adverse selection among early adopters—consumers who buy despite lower initial valuations tend to impose higher costs—which I derive from a model of learning and sorting into a new product. Stronger early selection creates a dynamic free-rider problem that can unravel markets before maturity, even when mature equilibrium is efficient. Market power mitigates unraveling through intertemporal cross-subsidization. Hand-collected deferred income annuity prices show evidence consistent with the mechanism’s central prediction, adverse selection falls as the market matures for groups with greater private information.

Keywords: Adverse selection, Insurance, Annuities; JEL Codes: D82, G22, D14

*Department of Economics, University of Western Ontario and Institute for Fiscal Studies. Address: Department of Economics, Social Science Centre, University of Western Ontario, London, Ontario, N6A 5C2, Canada. Email: rmcgee4@uwo.ca. Website: <https://sites.google.com/view/rorymcgee/home>. I thank Roy Allen, Eric French, Nirav Mehta, Sergio Ocampo, Martin O’Connell, Cormac O’Dea, Jesse Perla, and Lisa Tarquinio for helpful comments and suggestions. Ian O’Donnell provided excellent research assistance. This research was supported by the University of Western Ontario Faculty Research Development Fund.

1. Introduction

Selection markets—where consumers differ in the costs they impose on firms—are ubiquitous across annuities, insurance, credit, securities, used goods, and labor markets (Einav et al. 2021). These markets often exhibit low utilization, which empirical work routinely attributes to weak demand.¹ Yet, this explanation overlooks a puzzle: suppliers’ failure to offer more attractive products despite unmet demand. I show how adverse selection in new markets exacerbates demand shortfalls and drives unraveling.

This paper’s central insight is that adverse selection in immature markets creates a novel dynamic free-rider problem. When early adopters are more adversely selected, markets may fail to mature because initial entrants bear the costs of adverse selection while subsequent entrants compete away long-run rents, preventing the cross-subsidization of initial losses. Consequently, initial adverse selection can cause permanent unraveling even when the mature market would be efficient.

Whether early adopters are in fact more adversely selected is central, but is theoretically ambiguous. I give sufficient conditions for it and, rather than imposing them on the joint distribution of costs and willingness-to-pay directly, showing how they can be derived from primitives governing how consumers adopt a new product. In immature markets, potential consumers have less confidence in products and derive lower benefits from adoption. As demand falls, the consumers who still buy—those willing to adopt despite lower initial valuations—tend to be those imposing the highest costs on firms, so adverse selection is higher in immature markets. New evidence from the maturing deferred income annuity market supports this channel.

Using hand-collected annuity price data, I exploit age-based differences in available private information to disentangle the selection component of pricing from other dynamics common across purchase ages. I validate the key identifying assumption—that private information increases with purchase age—using subjective mortality beliefs from the Health and Retirement Study, showing that older respondents’ beliefs are significantly more predictive of realized mortality even conditional on health observables. Consistent with the model’s central prediction, the decline in measured selection as this market matures is larger for groups with more private information. I interpret this as evidence for the mechanism’s premise—that early adopters are more adversely selected—rather than as a direct test of the dynamic

¹See Lockwood (2012) and Pashchenko (2013) for annuities, Lockwood (2018) in long-term care insurance, and Nakajima and Telyukova (2017) and Cocco and Lopes (2020) in reverse mortgage markets. Alternatively see Einav et al. (2010) for results emphasizing adverse selection.

unraveling, which by its nature concerns markets that never form and cannot be observed. Yet, this decline is concentrated at the market’s transition to scale, a window that coincided with insurer entry and a change in broker fiduciary duty. I therefore stop short of attributing the aggregate decline to scale, relying instead on differential trends across purchase ages.

More broadly, the framework offers a joint explanation for three otherwise puzzling regularities in selection markets: low take-up of existing products, sluggish demand for new ones, and market inactivity despite substantial unmet demand. For example, [Zinman \(2014\)](#) highlights missing rungs in the consumer lending ladder, [Ameriks et al. \(2016\)](#) documents unmet demand for better long-term care insurance, and [Cocco and Lopes \(2015\)](#) establish demand for alternative reverse mortgage products. [Michaud and Amour \(2023\)](#) document potential demand for bundled risk management products combining existing insurance. Yet take-up remains very low: [Nakajima and Telyukova \(2017\)](#) and [Webb \(2011\)](#) document poor adoption of novel reverse mortgage and annuity products, respectively. Similarly, despite being available since 1954, variable annuities (bundling tax-favorable saving, insurance, and indexed options) remained a small market until the early 1990s ([Brown and Poterba 2006](#)).²

The paper makes three contributions. First, I develop general comparative-statics results for selection markets and microfound the sorting of early adopters—and hence their increased adverse selection—from primitives on product adoption, generating the payoffs behind the dynamic free-rider problem. Since monotonicity is not guaranteed with multidimensional heterogeneity ([Fang and Wu 2018](#)), I give sufficient, economically interpretable conditions in terms of the copula linking costs and willingness-to-pay. These conditions accommodate commonly used empirical specifications, extend the graphical framework of [Einav and Finkelstein \(2011\)](#) to cross-market and dynamic comparisons by characterizing demand and cost curve shifts, and deliver a partial ordering of equilibrium prices, quantities, selection, and welfare in terms of model primitives.

Second, I outline the novel free-rider problem and its implications in an intuitive framework. A key insight is that even would-be efficient markets can unravel during immaturity—a new, dynamic source of market failure building on [Akerlof \(1970\)](#). This challenges standard intuitions about market power and adverse selection. Static analyses, such as [Starc \(2014\)](#) and [Mahoney and Weyl \(2017\)](#), highlight that adverse

²Concern about missing insurance innovation dates back over half a century: [Rudelius and Wood \(1970\)](#) identify heterogeneous adoption of life insurance innovations, and [Dorfman \(1971\)](#) provides evidence of research activity and barriers to innovation, and [Murray \(1976\)](#) identifies innovations.

selection can mitigate market power’s negative impacts by constraining rent extraction, though market power exacerbates under-supply. In a dynamic setting, however, market power can alleviate these dynamic inefficiencies by allowing monopolists to cross-subsidize early losses with longer-run gains. This points toward a new rationale for protecting early market power, through patents or temporary regulatory barriers, rather than the standard prescription of restricting it, though the case for any specific instrument depends on the planner’s ability to target markets that will mature.

Third, although the dynamics of markets that never form cannot be observed directly, I show that a central empirical implication of the model holds in the market for deferred income annuities. I use a novel identification strategy exploiting age-based differences in the potential for private information to test whether adverse selection changes as the market matures. I find that the decline in adverse selection as the market matures is steeper for older purchasers, whom I show possess the most private information. This evidence provides empirical support for the mechanism’s premise, while leaving the non-existence of markets itself, necessarily, untested.

All three contributions build on a standard insurance demand model with multidimensional heterogeneity. Consumers are heterogeneous across costs, risk premia (which determine insurance surplus), and other dimensions capturing frictions or departures from rationality. I consider adverse selection settings where consumers’ willingness-to-pay correlates positively with their costs to insurers.³ To study unraveling in novel markets, I distinguish mature markets (well understood by consumers) from immature markets (new and unfamiliar), allowing consumers’ willingness-to-pay to vary with market maturity.

These differences in components of willingness-to-pay capture objective factors, such as counterparty risk (Briggs et al. 2023) or self-insurance patterns consumers cannot quickly adjust, and subjective factors including distrust of new products, inertia (Handel 2013), or learning dynamics (Israel 2005). They lower demand and increase consumer sorting on costs, increasing adverse selection in immature markets and generating equilibrium under-insurance with lower adoption and higher costs.

Related Literature

I show that a novel free-rider problem manifests in the dynamics of adversely selected markets. This connects to Kong et al. (2024), who show the “cream-skimming” incentives generated by price competition can undermine entry and unravel adversely selected

³Analogous results can be obtained for advantageous selection.

markets—so that market power can improve welfare by preserving entry. I extend this insight to a dynamic setting in which the *timing* of competition matters for market development. Early entrants bear the costs of market development, including higher adverse selection, while later entrants compete away the resulting rents by entering only after the market matures. Market maturation is thus a non-excludable investment that early entrants provide and later entrants consume for free, so that adverse selection becomes a form of the appropriability problem in innovation (Nelson 1959; Arrow 1962) and the market may never mature. Like Kong et al., this implies a role for limiting competition; but where their analysis is static, the dynamic mechanism here calls for protecting early entrants’ *future* rents—via patents or temporary regulatory barriers—so that intertemporal cross-subsidization remains feasible. This cross-subsidization is related to, but distinct from, policies designed to mitigate selection (Geruso and Layton 2017, review these in health insurance markets).

To microfound these incentives, I develop comparative-static tools that give non-parametric conditions for ordering partial unraveling in Akerlof (1970) markets, conditions that map onto commonly used empirical specifications such as the constant-absolute-risk-aversion utility of Cohen and Einav (2007). Hendren (2013) provides no-trade theorems for Rothschild and Stiglitz (1976) markets, while Attar et al. (2021) establish necessary and sufficient no-trade conditions in adverse-selection markets.⁴ I extend these non-entry results to settings where adverse selection declines over time, showing how the dynamics can produce permanently missing markets even when static analysis predicts efficiency.

I study distributional changes in choice frictions similarly to Handel et al. (2019), providing comparative statics for non-local changes in both choice friction distributions and insurance value of contracts across markets. This builds a partial ordering of equilibrium prices, quantities, and welfare in terms of model primitives. Handel et al., however, study local reforms (including information interventions) that alter consumer’s choice frictions and provide sufficient statistics formulas for their welfare impacts. As in Handel et al. and Spinnewijn (2017), I show how changes to model primitives lead to the rotations and shifts to cost and demand that are studied in other contexts (e.g. Starc 2014; Mahoney and Weyl 2017).

⁴A sizable literature studies endogenous contract terms under adverse selection (e.g. Rothschild and Stiglitz 1976; Handel et al. 2015; Veiga and Weyl 2016; Azevedo and Gottlieb 2017) or screening with intensive and extensive margins (Geruso et al. 2023), which is not this paper’s focus.

2. Adverse Selection in Mature and Immature Insurance Markets

I develop a canonical model of insurance demand and use it to study how market maturity affects adverse selection. I present the framework under competitive pricing and then introduce differences between mature and immature markets that generate stronger adverse selection among early adopters (Sections 2.1-2.4). I subsequently, extend it to market power and product differentiation (Section 2.5), and embed the resulting payoffs in a two-period dynamic entry game (Section 3)

2.1. Demand, Supply, and Adverse Selection

I consider an [Akerlof \(1970\)](#) model of insurance in the style of [Einav et al. \(2010\)](#). A single perfectly competitive and risk-neutral insurer sells insurance to a unit continuum of heterogeneous individuals at a common price p . Contract terms are exogenous and common across all individuals.

Demand. Individuals are heterogeneous across risk type π_i which determines the expected cost to the insurer; their risk premium r_i ; and potential demand frictions ε_i . These are private information for individual i .⁵ An individual buys insurance if and only if their *subjective* willingness-to-pay, expressed in money-metric price units,

$$\omega_i \equiv u(\pi_i, r_i, \varepsilon_i) = u(\underbrace{\pi_i + r_i + \varepsilon_i}_{\text{insurance value}}), \quad (1)$$

exceeds the price of the insurance. In addition to their risk type, individual i 's risk premium, r_i , determines the surplus in the insurance market.

Following [Spinnewijn \(2017\)](#), individual choices depend on additively separable demand frictions, ε_i , which can distort insurance demand. This agnostic approach nests important features including borrowing constraints, inertia, subjective beliefs, and bounded rationality. These frictions are heterogeneous and can lead individuals to overvalue ($\varepsilon > 0$) or undervalue ($\varepsilon < 0$) the insurance contract. Since risk premia and frictions have identical effects on consumer choices, I focus on perceived risk premia net of behavioral influences $\tilde{r}_i \equiv r_i + \varepsilon_i$, for the majority of the analysis.

⁵Conditioning on observable characteristics correlated with π_i may be illegal. Nevertheless, this can be interpreted as a model of insurance contracts in each partition of the observable characteristic space. This rules out screening contracts—changes to the screen may be important in immature markets.

Individual demand is given by $\mathbb{1}[\omega > p]$ which yields aggregate demand

$$D(p) = 1 - F_\omega(p) = E[\mathbb{1}[\omega > p]], \quad (2)$$

where F_x denotes the CDF of variable x with mean, μ_x , and variance, σ_x^2 .

Supply. The cost of providing insurance at price p is determined by the risk types of individuals purchasing insurance at that price.⁶ Consequently, *average* and *marginal* costs are given by:

$$AC(p) = E(\pi | \omega \geq p) \quad \text{and} \quad MC(p) = E(\pi | \omega = p). \quad (3)$$

When willingness-to-pay, ω , increases in insurer cost, π , there is adverse selection; marginal costs increase in price and the MC curve is downward sloping. Only the component of willingness-to-pay that correlates with cost generates adverse selection: risk premia and demand frictions orthogonal to cost shift demand without sorting buyers on cost. The empirical content concerns this cost-correlated component alone.

Equilibrium. Assuming continuous, monotonic, single-crossing demand and cost curves ensures a unique equilibrium if it exists. The equilibrium price is given by the break even price that pools expected costs

$$p^* : p^* = AC(p^*). \quad (4)$$

Consequently, the equilibrium allocation may not be efficient. An efficient allocation insures everyone whose willingness-to-pay exceeds their expected cost ($\omega_i > \pi_i$), those for whom the demand curve lies above the marginal cost curve (Panel A, Figure 1).

Furthermore, a positive equilibrium price does not necessarily exist. The insurance market unravels (Akerlof 1970) when individuals are not willing to purchase insurance at the pooled cost of higher-demand individuals. This occurs when the demand curve lies entirely below the average cost curve (Panel B, Figure 1 and Hendren 2013).

2.2. Mature and Immature Markets

I use superscripts *MM* and *IM* to denote the mature and immature markets and the relation $X \preceq Y$ to denote first order stochastic dominance. To make markets comparable,

⁶I abstract from administrative loadings or fixed costs which don't affect the main result.

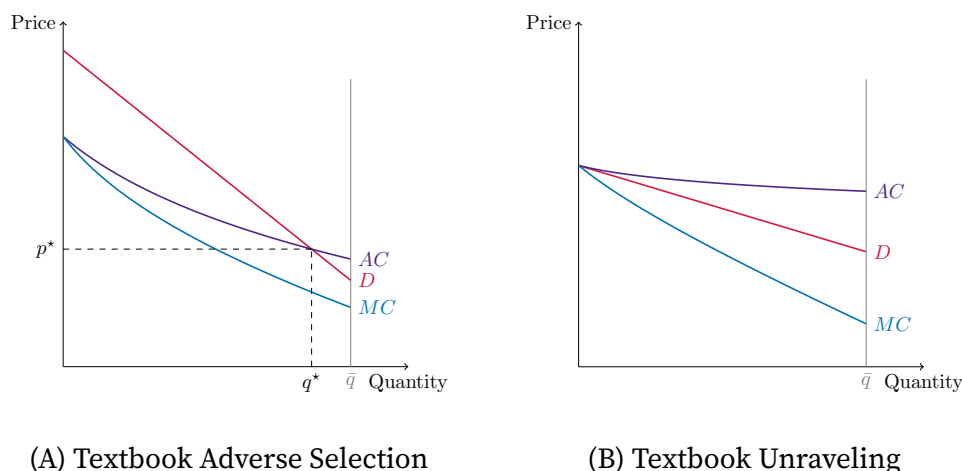


FIGURE 1. Equilibrium in the Insurance Market Without Demand Frictions

I hold the distribution of potential and realized risks fixed across mature and immature markets so that the product and risk pools are identical.⁷

ASSUMPTION 1. *Risk types, π_i , have identical marginal distributions in mature and immature markets.*

I then assume willingness-to-pay is depressed in immature markets, thus demand contracts. This can be driven by objective risk premia, r , or from demand frictions, ε . I begin by motivating differences in the objective component of net premia, $\tilde{r}$.

Three forces generate objective differences between risk premia in immature versus mature markets. First, portfolios may not yet be optimized to take advantage of a new product (e.g., savings used for self-insurance crowding out formal insurance), so the benefits of insurance rise as other choices adjust. Second, there may be greater objective uncertainty about coverage, hassle costs, or reclassification risk; in health insurance, for example, coverage of new drugs/procedures is uncertain and heterogeneous, which depresses premia. This uncertainty is likely larger for new types of insurance and lowers the risk premia. Third, counterparty risk may be higher when insurers collect premia now but only begin paying out far in the future (e.g., life insurance, deferred annuities, long-term care), distorting insurers' portfolio choice (Thompson 2010, models endogenous counterparty risk) and raising the likelihood of default or non-payment—again lowering risk premia.

Alternatively, lower demand may reflect more negative demand frictions in immature markets: status quo bias, distrust of new products, inertia, lack of awareness

⁷Thus neither firms nor individuals can alter the risk distribution in mature markets.

(Boyer et al. 2020), or narrow consideration sets. Frictions that dissipate with learning or peer experience (Chetty et al. 2013; Duflo and Saez 2003) fit the same representation. In short, immature markets have *more negative* demand frictions than mature markets, where more consumers have prior exposure (peers, advertising, past purchase).

ASSUMPTION 2. *Willingness-to-pay is on average larger and demand expands in mature markets: $\omega^{MM} \succeq \omega^{IM}$*

This formalizes how shifts in risk premia and demand frictions affect willingness-to-pay. Instead of demand frictions shifting systematically, consumers may have noisier beliefs about new products, reflected in greater dispersion of demand frictions capturing unbiased but diffuse priors. Assumption 2 is weaker than requiring stochastic dominance of objective risk premia, r , or demand frictions, ε . It only needs to be satisfied *on net*. Unlike Spinnewijn (2017), this does not require consumers are unbiased on average; systematic optimism or pessimism is consistent with Assumption 2 (see, e.g. Brown et al. 2021; O’Dea and Sturrock 2023).

Assumptions 1 and 2 restrict how marginal distributions differ between mature and immature markets. Yet, they do not capture how the sorting of consumers may change as markets mature. A natural primitive is that, in immature markets, cost-independent sources of willingness-to-pay heterogeneity are compressed. This strengthens the dependence between π_i and ω_i . I formalize this as a concordance ordering on the copula between π_i and ω_i :

ASSUMPTION 3 (Need-Based Early Adoption). *Resorting is such that copulas C , describing dependence between π_i and ω_i , satisfy the following concordance ordering (Tchen 1980): $C^{IM}(u, v) \geq C^{MM}(u, v) \forall u, v \in [0, 1]$*

This orders joint distributions *between* markets by implied dependence. Costs and willingness-to-pay have larger concordance—their joint density is more concentrated around a monotone graph.⁸

When consumers must expend resources to learn about a new contract or integrate it into an existing financial plan, those with the strongest insurance motive—high- π types—are disproportionately likely to adopt early, so the resulting pool of buyers is tightly sorted on insurer cost. As the market matures, adoption extends to consumers whose willingness-to-pay reflects portfolio complementarities, long-term planning, peer experience, or product-specific preferences rather than only expected cost. The

⁸See Appendix A.4 for a construction from assumptions on π_i and $\tilde{r}_i$ directly.

empirical evidence from deferred income annuities is consistent with this mechanism, but the framework does not require that it hold in *every* new market.

2.2.1. A Behavioral Microfoundation for Need-Based Early Adoption

Assumption 3 orders the dependence between costs and willingness-to-pay across maturity. This assumption simplifies the exposition but should be read as a reduced form. It can be explicitly microfounded, for example, when accumulated market experience lowers the cost of perceiving cost-orthogonal product value, so effective demand decouples from cost as the market matures.

First, decompose $\tilde{r}_i = h(\pi_i) + v_i$ with $v_i \perp \pi_i$ an idiosyncratic private value. Since u is strictly increasing and copulas are invariant to monotone marginal transforms, the copula of (π_i, ω_i) equals that of $(\pi_i, g(\pi_i) + v_i)$, with $g(\pi) \equiv \pi + h(\pi)$ defining the mapping from cost to willingness-to-pay with positive derivative $g' > 0$. In this case, the content of Assumption 3 concerns how cost-orthogonal private value v_i enters demand.

Additionally, assume consumers know the salient, cost-correlated motive $g(\pi_i)$ but perceive cost-orthogonal private value $v_i \sim N(0, \sigma_v^2)$ only through a signal,

$$s_i = v_i + \zeta_i, \quad \zeta_i \sim N(0, \sigma_\zeta^2(S)), \quad \zeta_i \perp (\pi_i, v_i),$$

where the precision of the signal increases in the stock of public product experience S , $d\sigma_\zeta^2(S)/dS < 0$, capturing peers, advisors, standardized contracts, or accumulated evidence among other margins. Effective willingness-to-pay is $\omega_i(S) = u(g(\pi_i) + \hat{v}_i(S) + \mu(S))$, where

$$\hat{v}_i(S) = \mathbb{E}[v_i \mid s_i] = a(S) s_i, \quad a(S) = \frac{\sigma_v^2}{\sigma_v^2 + \sigma_\zeta^2(S)} \in [0, 1], \quad (5)$$

and $\mu(S)$, $d\mu(S)/dS > 0$, captures the trust or familiarity channel of Section 2.2. Maturity is produced by the stock of past adoption, $S_{t+1} = (1 - \kappa)S_t + \phi q_t$, and is endogenous, but exogenous to any current consumer. When S is small the product is opaque, $\hat{v} \approx 0$, and purchase is driven by cost; as S grows, idiosyncratic value decouples demand from cost.

PROPOSITION 1. *Let $S^{IM} < S^{MM}$, g affine, and (π_i, v_i) jointly Gaussian. Then the copula C^S of $(\pi_i, \omega_i(S))$ is invariant to u and to the location channel $\mu(S)$, depends on S only through $\text{Var}(\hat{v}(S))$, and is strictly decreasing in S in the concordance order. Hence $C^{IM}(u, v) \geq C^{MM}(u, v)$ for all $(u, v) \in [0, 1]^2$, which is Assumption 3.*

The formal proof is in Appendix A. The proposition isolates the dependence channel—the substantive content of need-based early adoption—whereby rising signal precision concentrates the joint distribution of cost and willingness-to-pay around a monotone graph, independently of the marginals, of u , and of the location $\mu(S)$. The same location channel delivers the demand expansion of Assumption 2, which Lemma A1 in Appendix A establishes on the relevant range of equilibrium prices.

2.3. Equilibrium in Immature Markets

I use the graphical framework of Einav et al. (2010) (see also Einav and Finkelstein 2011, 2023) to characterize equilibrium differences between mature and immature markets. For each result, I provide closed-form proofs under joint normality and linear utility; Section 2.4 gives general sufficient conditions for monotonicity and within-market results under arbitrary marginals, with full proofs of general results in Appendix A.4.⁹

ASSUMPTION 4 (Joint Normality). π_i and $\tilde{r}_i$ are jointly normally distributed. Additionally, the variance of willingness-to-pay is constant, $\sigma_\omega^{IM} = \sigma_\omega^{MM}$, and correlation between risk type and willingness-to-pay is weaker in mature markets, $\text{corr}(\pi, \omega^{IM}) \geq \text{corr}(\pi, \omega^{MM}) > 0$.

Under joint normality, π and $\tilde{r}$ are multivariate normal with correlation $\rho_{\pi, \tilde{r}}$, and the joint distribution of cost type and willingness-to-pay is given by

$$\begin{pmatrix} \pi_i \\ \omega_i \end{pmatrix} \sim N \left[\begin{pmatrix} \mu_\pi \\ \mu_\omega \end{pmatrix}, \begin{pmatrix} \sigma_\pi^2 & \rho \sigma_\pi \sigma_\omega \\ \rho \sigma_\pi \sigma_\omega & \sigma_\omega^2 \end{pmatrix} \right],$$

where $\rho = \frac{\sigma_\pi + \rho_{\pi, \tilde{r}} \sigma_{\tilde{r}}}{\sigma_\omega} > 0$. The marginal cost curve is

$$MC(p) = E(\pi | \omega = p) = \mu_\pi + \rho \frac{\sigma_\pi}{\sigma_\omega} (p - \mu_\omega), \quad (6)$$

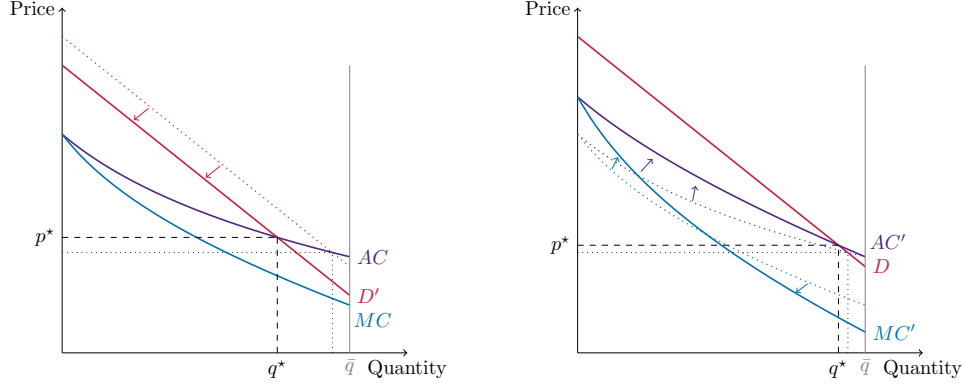
which is increasing in p when $\rho_{\pi, \tilde{r}}$ (and thus ρ) is positive. Average costs are

$$AC(p) = E(\pi | \omega \geq p) = \mu_\pi + \rho \sigma_\pi \frac{\phi(\frac{p - \mu_\omega}{\sigma_\omega})}{1 - \Phi(\frac{p - \mu_\omega}{\sigma_\omega})}, \quad (7)$$

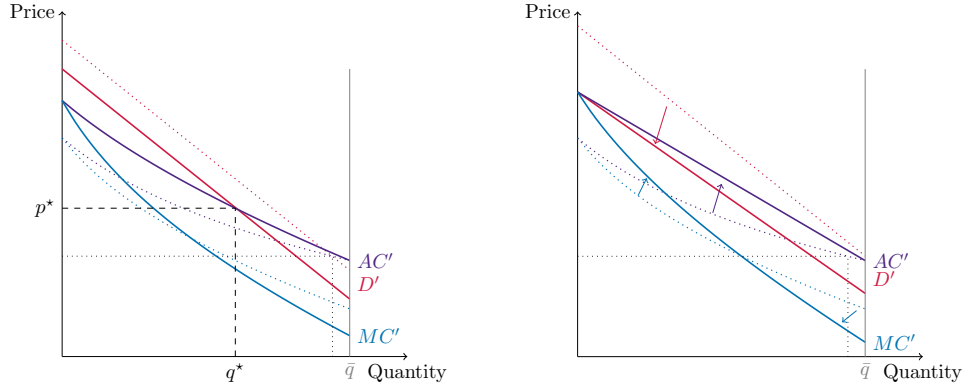
⁹Monotonicity of demand and cost curves does not require choice frictions are independent of other characteristics (Solomon 2023 finds evidence in favor of dependence). Consequently, choice frictions are a reduced form compatible with a range of behavioral microfoundations including rational inattention models and correlated mistakes affecting both risk premia and purchase decisions (Leive et al. 2022).

an endogenously truncated normal distribution, increasing in p for positive $\rho_{\pi, \tilde{r}}$.¹⁰

Demand Contraction. The implication of Assumption 2 for demand is immediate: at any given price, average willingness-to-pay is higher in mature markets, so more individuals purchase insurance. Panel A of Figure 2 depicts this *level effect* on demand: $D^{IM}(p) \leq D^{MM}(p) \forall p$.



(A) Demand Contraction (Level Effect) (B) Greater Adverse Selection (Sorting)



(C) Under-insurance in the Short Run (D) Unraveling in the Short Run

FIGURE 2. The Long and Short Run of Insurance Markets

This level effect holds average and marginal costs fixed at any given quantity. Although households require lower prices to induce purchase, willingness-to-pay

¹⁰Assumption 4 rules out changes in the dispersion of willingness-to-pay across mature and immature markets. Under this condition, changes to the primitives take the form of location shifts for the marginal distributions. This is weaker than the stochastic ordering of normals as it does not impose a common variance across mature and immature markets for π_i and $\tilde{r}_i$.

ordering is unchanged and the identity of marginal (and inframarginal) buyers at a given quantity is the same, thus equilibrium price rises while quantity falls (Panel A).

Cost Expansion. A pure level effect can raise average costs at a given price because it changes equilibrium quantity, but a change in sorting is required to shift cost curves *conditional on quantity*. When selection on cost increases, insurers face steeper marginal costs and higher average costs.

Order individuals by willingness-to-pay. Without re-sorting, this ordering is identical in mature and immature markets. Thus if q individuals buy insurance, the marginal consumer is the same and imposes identical expected costs on the insurer. With re-sorting, identities change: increased sorting on cost in immature markets implies that purchasing insurance signals higher average costs than in maturity.

Under normality, the decomposition of the difference in average costs makes this precise. Comparing across markets gives, $\forall p$:

$$\frac{AC^{IM}(p) - AC^{MM}(p)}{\sigma_\pi \rho^{IM}} = \underbrace{\left(\frac{\phi(\frac{p - \mu_\omega^{IM}}{\sigma_\omega})}{1 - \Phi(\frac{p - \mu_\omega^{IM}}{\sigma_\omega})} - \frac{\phi(\frac{p - \mu_\omega^{MM}}{\sigma_\omega})}{1 - \Phi(\frac{p - \mu_\omega^{MM}}{\sigma_\omega})} \right)}_{\text{level effect} \geq 0} + \underbrace{\left(1 - \frac{\rho^{MM}}{\rho^{IM}} \right) \frac{\phi(\frac{p - \mu_\omega^{MM}}{\sigma_\omega})}{1 - \Phi(\frac{p - \mu_\omega^{MM}}{\sigma_\omega})}}_{\text{re-sorting effect} \geq 0} \geq 0 \quad (8)$$

The inverse Mills ratios and their difference are always positive by the log concavity of the normal distribution, ensuring non-negativity. Both effects work in the same direction—raising average costs at a given price—as additional selection on costs among inframarginal consumers dominates the ambiguous effect on the marginal consumer.

The corresponding decomposition for marginal costs is:

$$MC^{IM}(p) - MC^{MM}(p) = \frac{\sigma_\pi}{\sigma_\omega} \rho^{IM} \left[\underbrace{(\mu_\omega^{MM} - \mu_\omega^{IM})}_{\text{level effect} \geq 0} + \underbrace{\left(1 - \frac{\rho^{MM}}{\rho^{IM}} \right) (p - \mu_\omega^{MM})}_{\text{re-sorting effect}} \right]. \quad (9)$$

Absent re-sorting, the increase depends only on the demand contraction. With re-sorting, the sign is ambiguous: raising the difference at prices above μ_ω^{MM} but lowering below, as the most adversely selected consumers are already out of the low-price pool.

Panel B of Figure 2 shows the outward expansion of the average cost curve in

immature markets.¹¹ Even holding the level of demand fixed, increased adverse selection among inframarginal consumers raises expected costs. Insurers set higher prices to cover these costs, despite some lower-cost buyers exiting.¹²

Substituting inverse demand into (7) gives the decomposition at a given quantity:

$$AC^{IM}(q) - AC^{MM}(q) = \underbrace{\rho^{IM} \sigma_{\pi} \left(1 - \frac{\rho^{MM}}{\rho^{IM}} \right)}_{\text{re-sorting effect}} \frac{\phi(\Phi^{-1}(1-q))}{q} \geq 0 \quad \forall q. \quad (10)$$

At a given quantity, differences in average costs depend only on re-sorting. Marginal costs rotate around the median ($MC^{IM}(q) \geq MC^{MM}(q)$ for all $q \leq 1/2$ and vice versa), so the adverse selection wedge is higher at any quantity demanded: $|AC^{IM}(q) - MC^{IM}(q)| \geq |AC^{MM}(q) - MC^{MM}(q)|$.

Partial Unraveling. Demand contraction ($D^{IM}(p) \leq D^{MM}(p)$) together with cost expansion ($AC^{IM}(p) \geq AC^{MM}(p)$) jointly establish:

PROPOSITION 2 (Partial Unraveling). *If an equilibrium exists, the equilibrium price is higher in immature markets and fewer consumers are insured, $q^{MM} \geq q^{IM}$ and $p^{MM} \leq p^{IM}$.*

PROOF. I first establish underinsurance in equilibrium. Let q^{MM} be the equilibrium quantity in the mature market and let $\tilde{D}(q) = D^{-1}(q)$ denote inverse demand. Then evaluating at mature market equilibrium quantity

$$\tilde{D}^{IM}(q^{MM}) \leq \tilde{D}^{MM}(q^{MM}) = AC^{MM}(q^{MM}) \leq AC^{IM}(q^{MM}), \quad (11)$$

where the first inequality is the (inverted) demand contraction and the second inequality is the cost-expansion. This establishes disequilibrium, $\tilde{D}^{IM}(q^{MM}) \leq AC^{IM}(q^{MM})$, which is strict unless the immature and mature markets are identical. Thus if inverse-demand and average cost intersect it must be at a lower quantity: $q^{IM} < q^{MM}$.

Using monotonicity of the average cost curve and that it is decreasing in q establishes

$$p^{IM} = AC^{IM}(q^{IM}) \geq AC^{IM}(q^{MM}) \geq AC^{MM}(q^{MM}) = p^{MM}. \quad (12)$$

¹¹This is a rotation around full insurance as in Mahoney and Weyl (2017). The difference in average costs in (10) is decreasing in quantity, reaching zero at full coverage: $AC^{IM}(1) = AC^{MM}(1) = \mu_{\pi}$. In contrast, Starc (2014) considers rotation around equilibrium demand.

¹²This can occur even if marginal costs in immature markets lie below demand which, absent choice frictions, would make full coverage efficient.

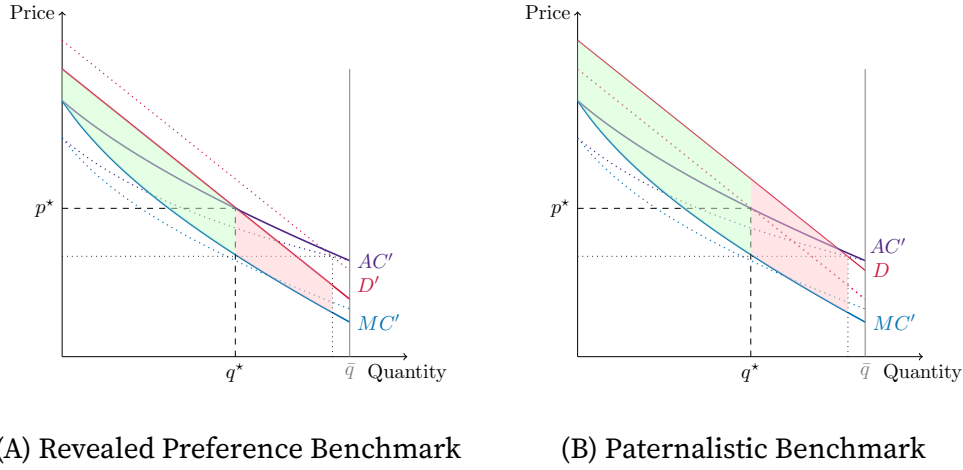


FIGURE 3. Welfare Losses in Immature Insurance Markets

Furthermore, if there exists a price $p^{IM} \neq p^{MM}$ that clears the immature market then differences across markets satisfy *strict dominance*. $\square$

Panel C of Figure 2 illustrates the new equilibrium. Marginal costs lie below demand in both mature and immature markets, so full insurance is efficient in both. Nevertheless, relative to the mature market there is additional under-insurance due to both the level effect and increased adverse selection in the immature market. The equilibrium price increase and quantity reduction exceed either effect in isolation (Panels A and B).

Immaturity can also cause complete unraveling:

COROLLARY 1 (Complete Unraveling). *Would-be efficient mature markets can unravel before they reach maturity.*

This happens when the average cost curve lies above demand for all prices (Panel D of Figure 2). Thus, even a market that would be efficient if it could mature can unravel in its early stages. As Section 3 shows, this can lead to permanently missing markets.

Under-insurance ($q^{IM} < q^{MM}$) does not depend on whether lower willingness-to-pay is driven by risk premia or choice frictions. Computing welfare costs, however, requires a view on their relative importance. The next proposition evaluates welfare under two policy-relevant benchmarks, illustrated in Figure 3.

PROPOSITION 3 (Efficiency Costs). *In the case of partial unraveling (Panel C), where $q^{MM} \geq q^{IM}$, consumer surplus in immature markets is lower at the immature market equilibrium than the mature market equilibrium ($CS^{IM}(q^{IM}) \leq CS^{IM}(q^{MM})$). This is true irrespective of whether demand for insurance is evaluated at:*

i. Willingness-to-pay in immature markets (revealed preference benchmark)

ii. Willingness-to-pay in mature markets (paternalistic benchmark)

Thus, the equilibrium in immature markets is pareto dominated by the mature market.

PROOF. Firms earn zero profits in equilibrium, so consumer surplus orders equilibria. Under the revealed preference benchmark, the difference in consumer surplus is

$$CS^{IM}(q^{MM}) - CS^{IM}(q^{IM}) = \int_{q^{IM}}^{q^{MM}} \tilde{D}^{IM}(q) - MC^{IM}(q) dq \geq 0, \quad (13)$$

where $\tilde{D}(q) = D^{-1}(q)$ and non-negativity follows from $q^{MM} \geq q^{IM}$ (Proposition 2). Under the paternalistic benchmark,

$$CS^{IM}(q^{MM}) - CS^{IM}(q^{IM}) = \underbrace{\int_{q^{IM}}^{q^{MM}} \tilde{D}^{MM}(q) - \tilde{D}^{IM}(q) dq}_{\text{Paternalism } (\geq 0)} + \underbrace{\int_{q^{IM}}^{q^{MM}} \tilde{D}^{IM}(q) - MC^{IM}(q) dq}_{\text{Case (i)} (\geq 0)}, \quad (14)$$

where the paternalistic contribution is positive by demand contraction.¹³ $\square$

Under the revealed preference benchmark, this requires $q^{MM} \geq q^{IM}$, which holds when differences across markets reflect fundamentals rather than frictions inflating mature demand. The paternalistic benchmark (case ii) always holds, since it evaluates immature surplus at mature preferences, where q^{MM} is by construction efficient.

2.4. General Sufficient Conditions

The preceding results are established under normality. With multidimensional heterogeneity, the critical assumption for adverse selection is the dependence between individual willingness-to-pay and costs (Fang and Wu 2018), which Assumption 3 pins down across markets. I use a copula approach to decouple dependence from marginal distributions, allowing the analysis to extend flexibly beyond stylized examples.

The following proposition formalizes familiar sufficient conditions for within-market monotonicity. It relies on the monotone likelihood ratio property (Milgrom 1981), which Appendix A shows to be stronger than necessary.

¹³It follows that consumer surplus in mature markets is generically larger than in immature markets.

PROPOSITION 4 (Adversely Selected Markets). *Let the density $f(\cdot)$ satisfy the Monotone Likelihood Ratio Property (MLRP), i.e. $f(\omega_1|\pi_1)/f(\omega_1|\pi_2) \geq f(\omega_2|\pi_1)/f(\omega_2|\pi_2)$ for any $\omega_1 \geq \omega_2$ and $\pi_1 \geq \pi_2$. Then:*

- i. *The market is adversely selected and marginal costs, $MC(p)$, are increasing in price.*
- ii. *Average costs, $AC(p)$, are monotonically increasing in price.*
- iii. *The adverse selection wedge, $AC(p) - MC(p)$, is weakly positive (strict under strict MLRP).*

The formal proof is given in Appendix A. Intuitively, MLRP allows the insurer to infer higher costs from the signal of higher willingness-to-pay. Crucially, this does not depend on marginal distributions. For a bivariate distribution with copula $C(u, v)$ and copula density $c(u, v)$, MLRP holds if and only if:

$$\frac{\partial^2}{\partial u \partial v} \log c(u, v) \geq 0. \quad (15)$$

This restricts the shape of the scale-invariant rank dependence *within* a market, formalizing that larger X values tend to be realized with larger Y values. It implies positive quadrant dependence and positive correlations and regression coefficients.¹⁴

Example. Consider the specification in [Cohen and Einav \(2007\)](#), where consumers have constant absolute risk aversion θ_i and face a loss m arriving with probability λ_i . Then $\tilde{r}_i = r_i = \frac{\ln(1 - \lambda_i + \lambda_i e^{\theta_i m})}{\theta_i} - \lambda_i m$, which is non-normal even when λ and θ are (log-)normally distributed. The copula conditions above, which impose no marginal distribution restrictions, extend the implication of the normality benchmark to such settings.

Under these general conditions and Assumptions 1–3, demand contractions and cost expansions hold for arbitrary marginal distributions. Propositions 2 and 3 then follow immediately, as their proofs depend only on monotonic demand and cost curves and not on a specific distribution. Formal statements and proofs are in Appendix A.4.

2.5. Differentiated Goods and Market Power

I now extend the model by relaxing product homogeneity and allow for market power. There are $j \in \{1, \dots, J\}$ firms offering identical insurance contracts bundled with

¹⁴As dependence is measured in ranks, the cross-partial derivative captures more than the correlation between π and ω : it encodes how jointly heavy-tailed they are. Appendix A.3 derives sufficient conditions directly on the conditional family $\tilde{r}|\pi$ — namely, that $\tilde{r}|\pi$ satisfies MLRP in $(\tilde{r}, \pi)$ and each conditional density is log-concave — that imply MLRP for the induced joint distribution of (ω, π) .

firm-specific differentiated attributes X_j . As in the equilibrium described above, firms maximize profits by setting a uniform price. Under Bertrand-Nash price competition, firm j 's equilibrium pricing equation (replacing equation 4) becomes:

$$p_j^* - AC(p_j^*) = \frac{1}{\eta_{D,p_j}} - \left(AC(p_j^*) - MC(p_j^*) \right), \quad (16)$$

where the second term captures pricing pressure from adverse selection. Rearranging yields the standard monopolist pricing. As in Mahoney and Weyl (2017), the impact of increased adverse selection depends on the quantities in the mature market:

LEMMA 1 (Under-Insurance with Product Differentiation). *Assume quantities in the mature market are small ($q_j^{MM} < \tilde{q}$) and the elasticity advantage in mature markets is bounded so that $\eta_{D,p_j}^{MM}/\eta_{D,p_j}^{IM} \geq 1 - \eta_{D,p_j}^{MM} [MC^{IM}(q_j) - MC^{MM}(q_j)]$ for $q_j < \tilde{q}$.¹⁵*

It follows that, if an equilibrium exists, equilibrium prices are higher and quantities lower in immature markets: $q_j^{MM} \geq q_j^{IM}$ and $p_j^{MM} \leq p_j^{IM}$.

This confirms that adverse selection reduces firms' incentive to supply differentiated products in immature markets, generating higher prices, lower quantities, reduced profits, and lower consumer surplus.¹⁶

Market power reduces but does not eliminate under-insurance in immature markets. The key question is whether a firm with market power can earn positive profits during immaturity—and if so, whether dynamic entry incentives can sustain the market through its early phase. I turn to this next.

3. Will Deep Pocketed Insurers Enter If Immature Markets Unravel?

Section 2.3 establishes that immature markets can suffer from worse adverse selection and may unravel even when the mature market would be efficient. I now ask whether dynamic incentives counteract these forces. Using a two-firm, two-period model, I show that adverse selection produces a novel form of dynamic free-rider problem that can prevent markets from ever reaching maturity.

¹⁵This rules out cases where mature-market demand is sufficiently inelastic that firms prefer to restrict quantities despite the reduction in costs.

¹⁶Corollary A4 in Appendix A.4 provides conditions guaranteeing this. These are satisfied under constant shape restrictions, e.g. stochastic dominance for log-concave willingness-to-pay distributions (Milgrom and Weber 1982; Chade and Schlee 2012; Chade and Swinkels 2021).

In each period, firms choose whether to offer insurance taking contract terms as given. They observe entry decisions and post prices, competing à la Bertrand, so prices equal average costs if both firms enter. If only one firm enters, they exploit market power and equate marginal revenue with marginal cost. The market begins as immature and updates to mature in the second period if and only if at least one firm enters in the first period and sells positive quantity.¹⁷

The following result, proof in Appendix B, characterizes how immaturity interacts with dynamic incentives and can endogenously create market incompleteness:

PROPOSITION 5. *If markets unravel and a monopolist cannot earn positive profits in immature markets, there exists a subgame perfect equilibrium where no firm enters. Moreover, this is the unique extensive-form trembling hand perfect equilibrium.*

Importantly, this depends only on immature market primitives and holds even when the mature market is efficient: static adverse selection in new markets can create missing mature markets regardless of mature market efficiency. The two-firm, two-period setup is chosen only to simplify exposition. The same insight survives in a general model in which firms cross-subsidize entry from future rents, the environment matures stochastically, and competition plays out over many periods. Non-entry holds whenever a lone entrant cannot earn positive profits by gambling on maturity, which limits the future rents available to cross-subsidize early losses. Relaxing Bertrand competition and allowing arbitrarily many firms delivers the same conclusion; more competition makes the market *less* likely to form. Even an arbitrarily well-capitalized entrant therefore stays out—the binding constraint is long-run profitability inclusive of rents, not capital. I provide results for the general case in Appendix B.

The intuition reveals a new free-riding problem preventing market maturation. Entering an immature market is an investment: the entrant bears negative profits while familiarizing consumers with the product, reducing future adverse selection for all firms. If monopolist profits in the mature market were guaranteed, absorbing these losses might be profitable. But competitors can wait and enter only after this investment is made, competing away the long-run rents that would otherwise cross-subsidize early losses. Consequently, no firm enters.¹⁸

This logic requires that no positive profits exist in the market's infancy. Unlike settings where low initial demand alone deters entry through scale economies, here

¹⁷Consumers learn about the generic "product" offered by firms instead of developing brand loyalty.

¹⁸This assumes aggregate state transitions only once contracts exceed a positive threshold, so firms cannot "enter" by charging prices guaranteeing zero profits and no sales.

the customer pool itself is unprofitable to serve: adverse selection makes the marginal entrant's expected costs exceed any feasible price. To see this, consider the standard profit margin expression from equation (16) (see, also [Starc 2014](#); [Kong et al. 2024](#)):

$$p - AC(p) = \frac{1}{\eta_{D,p}} - (AC(p) - MC(p)). \quad (17)$$

Firms may earn negative monopolist profits when the adverse selection wedge, $AC(p) - MC(p)$, exceeds the Lerner markup at all quantities.¹⁹ No price remedies this because higher prices shed the lowest-cost buyers first, leaving average costs weakly higher, so the demand curve lies everywhere below the average cost curve as in the complete unraveling case of Figure 2D.

3.1. Implications For Market Intervention

If mature markets generate surplus, maturation may be socially optimal despite no-trade in infancy. If the competitive allocation is efficient once the market matures, any intervention is optimally phased out as the primitives evolve; permanent solutions are generically inefficient. The first-best instrument is therefore a *targeted, time-varying* subsidy that offsets early losses and is withdrawn at maturity. Implementing it, however, requires precisely the information that is unavailable when markets are missing: the planner must know *which* nascent markets would mature into efficient ones, and absorb losses only in those. This informational requirement, rather than the principle of intervention, is the binding constraint.

This motivates asking what can be achieved by instruments that do *not* require the planner to identify maturing markets *ex ante*. Removing barriers to patenting, or lengthening patent protection, raises monopolist profits in immature markets by protecting early entrants against free-riders, and so acts as a decentralized time-varying subsidy that depends only on patent length.²⁰ A similar effect can arise from regulatory frictions on new products—disclosure or reporting mandates that slow cross-firm diffusion ([Benhabib et al. 2021](#)) and lengthen the period of market power.²¹

The tradeoff is that these instruments are blunt: requiring no information about

¹⁹[Kong et al.](#) similarly discuss entry under imperfect competition and derive a threshold condition on the slope of the average cost curve to sustain a given market structure.

²⁰See [Williams \(2017\)](#) for a survey of other considerations in optimal patent design. Patents additionally require information disclosure, which can help determine optimal patent length.

²¹Consistent with this channel, empirical work documents that regulation can suppress innovation ([Griffith and Macartney 2014](#); [Aghion et al. 2023](#)).

which markets will mature, cannot discriminate, and extend protection to every new product—including those that would, and arguably should, remain missing. Whether they improve welfare on net depends on the distribution of latent market types and the deadweight cost of market power in those that mature, a margin on which the model is silent. The analysis therefore identifies a *rationale* and a *direction*—that early market power can be welfare-improving in selection markets, reversing the static prescription to restrict it—rather than an unconditional case for any particular instrument.

The cross-subsidy that sustains entry need not be intertemporal. A firm with market power can absorb the cost of serving early adopters either by carrying losses forward within the product—trading early losses against later rents, as above—or by cross-subsidizing across products at a point in time, as when a dominant multi-product insurer prices a new contract below standalone cost and recovers the difference elsewhere in the relationship. Both require market power, and are instances of the same force: market power sustaining a selection market that would unravel.

Examples of Phased Intervention. Several insurance markets illustrate the first-best logic of temporary, targeted intervention withdrawn at maturity. The ACA’s temporary reinsurance and risk-corridor programs (2014–2016) channeled funds to plans with high-cost enrollees while the exchange risk pool stabilized, and sunset on schedule. The Dutch basic health insurance reform used ex post cost compensation to keep adverse selection from destabilizing the market at launch, then replaced it with permanent ex-ante risk-equalization as the data infrastructure matured.²² The Terrorism Risk Insurance Act (2002) introduced a government backstop after private terrorism reinsurance collapsed, with the public cost-share reduced over successive reauthorizations. Each calibrates intervention to an entry barrier the permanent market structure cannot yet resolve and then withdraws it—the structure the model assigns to time-varying policy. Each also relied on market-specific information to target and time support, the absence of such information motivates the decentralized alternatives above.

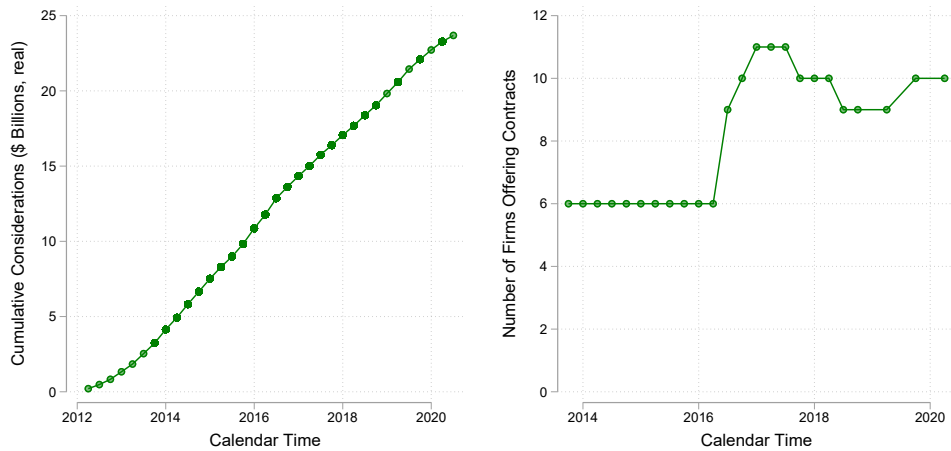
4. Evidence on Early Adopters in the Deferred Income Annuity Market

I now provide evidence that early adopters are more adversely selected. I study the comparison between different stages of market maturity and show how they align with

²²See [van de Ven and Schut \(2011\)](#) for an overview of the Dutch risk-equalization system and [McGuire et al. \(2020\)](#) for a comparative discussion.

testable implications in Section 2. Specifically, I focus on deferred income annuities—a product similar to a traditional annuity, but where payouts do not begin until a date (or age) agreed in the future. Hence, they provide deferred income.

Data on the aggregate size of the market, as measured by the dollar value of new annuity considerations (the amount paid to an insurance company to buy an annuity), is collected from the quarterly reports of the Life Insurance Marketing and Research Association.²³ Figure 4A shows how cumulative sales evolved over time. At the start of this period the market for deferred income annuities was small, totaling around \$200 million and accounting for less than 1% of the overall fixed income annuity market. The market expanded rapidly between 2012 and 2014, with the value of new considerations more than quadrupling in size. The deferred income annuity market is well suited to testing the mechanism precisely because it was genuinely immature at the start of the sample. Observing maturation of a selection market from inception is rare, and the modest initial scale is exactly what makes the mechanism testable in the data.



(A) Cumulative Deferred Income Annuities Considerations (B) Distinct Firms Operating in the Market for Deferred Income Annuities

Notes: Panel A plots cumulative new deferred income annuity considerations in real terms; Panel B the number of distinct insurers quoting deferred income annuities each quarter. Sources: LIMRA U.S. Individual Annuity Sales Survey (Panel A) and *Annuity Shopper* (Panel B).

FIGURE 4. Growth In The Deferred Income Annuities Market

To study the equilibrium relation between price, quantity, and selection I complement this data on aggregate quantities with hand-collected quotes for individual

²³Specifically, the U.S. Individual Annuity Sales Survey published by the LIMRA Secure Retirement Institute. Sales are not reported pre-2012 as the size of the deferred income annuities market is minimal.

annuity policies sourced from *Annuity Shopper* magazine between 2013 and 2020Q1.²⁴ This was a longstanding annuity and life insurance industry periodical, targeting potential annuitants, financial advisors and Human Resources professionals. Consequently, it is in the interests of life insurers to provide quotes on their products.²⁵

The sample starts in 2013Q3, when *Annuity Shopper* began collecting deferred-annuity quotes, just ahead of a major policy shift: the 2014 IRS/Treasury introduction of Qualifying Longevity Annuity Contracts (QLACs). By allowing deferred income annuities inside 401(k)s and IRAs and exempting qualifying premiums from required minimum distributions, QLACs likely helped expand demand, attract insurer entry, and alter buyer composition. Figure 4B shows how the number of firms offering deferred income annuities changes over time. Broadly, there are two regimes. First, a period where the number of entrants is stable at six distinct insurance firms. Second, a period averaging 10 firms. There are 13 firms providing deferred income annuity quotes over the whole sample. This expansion in the number of operating firms coincides with reforms to the fiduciary duty of brokers operating in brokerage markets (Egan et al. 2022). The Department for Labor’s fiduciary rule was proposed in 2015, finalized in 2016, and vacated in 2018. The strategy below therefore accounts for common time period effects in the market. Entrants are primarily large national insurers who infrequently exit product lines and the market at large, with New York Life the largest insurer.²⁶ I provide more details on the composition of entrants and their product mixes in Appendix C and return to the role of market power below.

In each period, national insurers provide quotes for a slate of policies, priced separately by gender (reflecting differences in survival risk), by the purchaser’s current age, and by the deferral age at which payouts begin. Quotes are available for three purchase ages (45, 55, and 65) and five deferral ages (65, 70, 75, 80, and 85), with up to 28 quotes per insurer in a period, reporting the annual payout per \$100,000 of premium. I collect a quarterly panel of 5,104 such quotes, focusing on single-life non-refundable policies without a guarantee period (Einav et al. 2010, study adverse selection in the

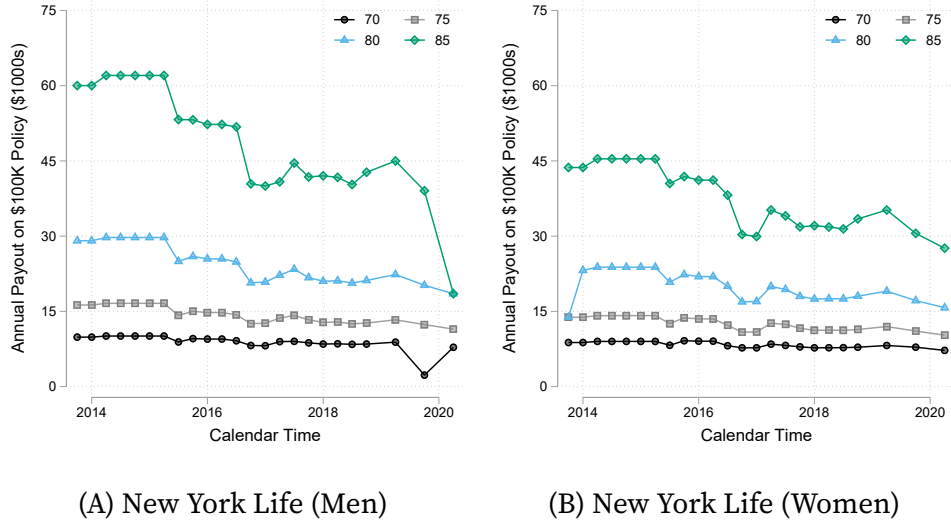
²⁴Previously The Annuity Shopper’s Buyers Guide. Before the third quarter of 2013 quotes for deferred income annuities are not available and, sadly for retirement researchers, it ceased to operate in 2021.

I exclude the COVID-19 period from the sample. The pandemic was a large age-dependent mortality shock, and large monetary-policy response that collapsed yields, disrupting annuity markets. Each may shift mortality expectations and pricing differentially across purchase ages—the margin the design exploits—threatening the assumption that non-selection factors evolve similarly across age groups. Results are qualitatively similar when these periods are included.

²⁵Indeed, across all products over 40 large US insurers provided quotes for a variety of products in the July 2015 issue. Appendix C provides an example testimonial.

²⁶Per their 2017 submission to the Department of Labor ([available here](#)) they had a 33% market share.

guarantee decision). A key advantage of the setting is that the product is plausibly homogeneous, unlike health insurance—a common setting for selection studies—where in- and out-of-network differences make products heterogeneous.



Notes: Annual payout per \$100,000 premium for single-life, non-refundable deferred income annuities purchased at age 65, by gender, and deferral age (70–85). From *Annuity Shopper*, 2013Q3–2020Q1.

FIGURE 5. Deferred Income Annuity Payouts at Purchase Age 65 By Insurer

Figure 5 shows payouts from annuities purchased at the age of 65 with policies offered by New York Life, the largest insurer in this market. The left shows policies for men and the right shows policies for women with each line corresponding to a deferral age from age 70 to 85. Two important facts emerge from this exercise. First, there are clear differences in policy generosity by gender. Second, there are systematic differences by the age at which the contract begins paying out. Policies with later deferral dates, which expose consumers to greater mortality risk, offer larger payouts on average and systematic differences across men and women are consistent with differences in their expected longevity. Additionally, there are small differences across insurers (Appendix C). Common market-level trend in payouts can reflect two forces. First, the opportunity cost of foregone investment (Mitchell et al. 1999), which therefore co-move with interest rates over the sample. Second, a changing annuitant pool which highlights the potential for selection to drive pricing decisions. The selection measure used in the analysis discounts by insurer cost of capital and so accounts for this common interest-rate trend.

4.1. Empirical Tests for Selection

I measure selection as it operates through the posted price schedule using the money's-worth tradition of [Finkelstein and Poterba \(2002\)](#). This differs from the conditional positive-correlation test of [Chiappori and Salanie \(2000\)](#) (see also, [Chiappori and Salanié 2013](#)), which asks whether coverage and risk correlate given the insurer's full information set. The available data preclude conditional correlation tests, but the price-based notion of selection matches the context. Insurers post a single schedule by age, gender, and deferral age, and the adverse-selection wedge at that price is what governs equilibrium quantity and efficiency.

In this context, the price based approach is also robust to a concern that complicates the conditional test—because selection is a property of the schedule, it is immaterial whether a buyer's risk is unknown to the insurer or known but unpriced.²⁷ What generates the *decline* in this wedge as the market matures is then not the insurer learning—the schedule conditions on age and gender throughout—but the demand-side resorting of Section 2.2.1. Adoption extends to cost-orthogonal buyers and dilutes the cost-willingness-to-pay correlation in the transacting pool. The strategy below isolates this by exploiting that it operates more strongly at older purchase ages, where there is more cost-relevant information outside the schedule.

4.2. Measuring Selection and Market Maturity

Each policy offered to annuitants at a purchase age, a , of gender, g , is indexed by the deferral age, d , at which payouts begin, the identity of the insurance firm, i , as well as the date, t , at which it is purchased. Each annuity, $\{a, d, g, i, t\}$, offers a fixed income A after the deferral age. I follow [Finkelstein and Poterba \(2002\)](#) and adopt a money's worth calculation to infer selection from annuity payouts. To do this, I calculate the expected present discounted value of the associated annuity income stream

$$EPDV_{a, d, g, i, t} = \frac{1}{\chi_a(g, t - a)} \sum_{h=a}^{110} \frac{A \times \mathbb{1}[h \geq d] \times \chi_h(g, t - a)}{\prod_{j=1}^{h-a} R_j(t)}, \quad (18)$$

where $\chi_h(g, t - a)$ is probability of surviving until age h as a function of annuitant gender and birth cohort $t - a$. The insurer's gross return at a horizon j from date t is $R_j(t)$. From

²⁷The empirical strategy below isolates fixed differences in the insurer's risk pools.

this I compute the money's worth value of the annuity (its implicit price):

$$MW_{a,d,g,i,t} = \frac{1}{100,000} EPDV_{a,d,g,i,t}. \quad (19)$$

A money's worth value less than one indicates that the annuity is expected to pay out less than the upfront cost and vice-versa. In the absence of annuity product-specific mortality tables (see [Finkelstein and Poterba 2004](#) for evidence of differential mortality across products) the money's worth value can be used to infer deviations from the mortality table used to calculate it. In this sense, it allows researchers to infer differential mortality and the presence of adverse selection (see also [Warshawsky 1988](#); [Poterba and Solomon 2021](#)).²⁸ I build a measure of selection using the per dollar deviation from the expected present discounted value under actuarially fair pricing using gender-specific population mortality tables.

To construct money's worth I use the BBB corporate bond yield at 10-year maturity to determine the insurer's interest rate, yielding gross return R_j , and cohort life tables for the age a population to determine mortality risk $\chi_a(g, t - a)$ at a given point in time (interpolating successive cohort life tables). Implied adverse selection is given by

$$S_{a,d,g,i,t} = 1 - MW_{a,d,g,i,t}, \quad (20)$$

where a larger positive value indicates greater adverse selection.

The testable implications of Section 2 relate adverse selection and market maturity. I proxy unobserved market maturity using the natural logarithm of cumulative sales in the market for deferred income annuities up to that point in time (Figure 4A), $\ln Q_t$. As the measure of new considerations used to construct cumulative sales is not available at a finer disaggregation I use total sales for all deferred income annuities, rather than sales at the characteristic level. This has an intuitive interpretation as a measure of broad market popularity that corresponds to maturity discussed in Section 2. Using a log transformation implies that there is a concave-increasing relationship between maturity and cumulative sales of deferred income annuities.

²⁸ [Cannon and Tonks \(2016\)](#) argue that prudent insurers may deviate from actuarially fair pricing when future mortality risk is uncertain. The empirical strategy below allows for this.

4.3. Empirical Strategy

A key threat to identifying the effect on adverse selection is that alternative strategic considerations (e.g., dynamic price discrimination) in maturing markets may generate observationally equivalent pricing patterns even absent adverse selection. Thus, the effect of maturity on pricing may not capture the effect of adverse selection.

Absent private information, it would not be possible for adverse selection to change as the market evolves. Therefore, to address these identification concerns and directly test the model's implications, I use differences in available private information by age to exploit variation in the potential for adverse selection. Specifically, as health inequality increases by purchase age (see, e.g., [Hosseini et al. 2022](#); [Russo et al. 2026](#)), private information about the future mortality of potential annuitants is also increasing. Equation 10 above formalizes this simple intuition which extends to the general model.

This motivates an empirical strategy that exploits differences in the effects of market maturity by purchase age. I leverage the differential effect of market maturity between those age 45 and those aged 55 and 65. The key identifying assumption is a form of parallel trends assumption: conditional on other controls strategic considerations, other pricing decisions, administrative costs, profit rates, and other price-relevant factors would evolve similarly with market maturity if 55- and 65-year-olds had the same level of mortality-relevant private information as 45-year-old consumers. This strategy is implemented by estimating the following specification²⁹

$$\ln S_{a,d,g,i,t} = \gamma_a + \gamma_d + \gamma_i + \gamma_g + \lambda_t + \delta \mathbb{1}_{[a>45]} \times \ln Q_t + u_{a,d,g,i,t}, \quad (21)$$

where γ_a , γ_d , and γ_g denote purchase-age, deferral-age, and gender fixed effects, and the time effects λ_t absorb every factor common to all purchase ages within a period—the common effect of market maturity, aggregate demand shocks, population-level mortality improvements, shifts in retirees' financial resources, and legislative changes including the broker fiduciary reform ([Egan et al. 2022](#)). The parameter of interest, δ , is therefore identified only from *differential* movements across age groups, it measures how the response of selection to market maturity differs for higher-private-information purchasers. The model predicts adverse selection falls as the market matures, and more steeply where private information is greater, since the common component is absorbed, this is $\delta < 0$. The design is akin to a parametric difference-in-differences, with

²⁹The log-specification excludes 46 observations from age-45 purchasers. Including these in a linear specification does not change the evolution of selection as I report in Appendix C.

ex-ante exposure (private information) playing the role of treatment and cumulative sales providing continuous variation in maturity.³⁰

The insurer fixed effects γ_i absorb compositional change from entry and exit. Insurers may serve different populations through heterogeneous geographic penetration or face different administrative costs. Absent this control, such differences would confound the effect of maturity.³¹

I also estimate a specification that splits the differential by purchase age:

$$\ln S_{a,d,g,i,t} = \gamma_a + \gamma_d + \gamma_i + \gamma_g + \lambda_t + \delta_{55} \mathbb{1}_{[a=55]} \ln Q_t + \delta_{65} \mathbb{1}_{[a=65]} \ln Q_t + u_{a,d,g,i,t}. \quad (23)$$

This provides a direct test of a second prediction—that markets with greater cost-relevant private information see larger reductions in adverse selection as they mature, $\delta_{65} < \delta_{55} < 0$ —and corroborates the identifying assumption, since the ordering by purchase age should track the ordering by private information.

For inference I two-way cluster by insurer and time period (Cameron et al. 2011), allowing insurer pricing errors to be serially correlated across quarters and permitting arbitrary contemporaneous correlation across insurers within a quarter. Because the number of clusters is modest, I report two-way wild cluster bootstrap p -values (Roodman et al. 2019; MacKinnon et al. 2021), bootstrapping at the insurer level.

4.4. Adverse Selection Among Early Adopters

Table 1 reports estimates of equation (21). The first column includes only purchase-age and time controls: the differential decline for higher-private-information purchasers is $\hat{\delta} = -0.322$, negative and statistically significant. Selection falls more as the market matures for the purchase ages with more private information, relative to age 45—the differential the model predicts. Figure 6 shows this is concentrated at the transition to scale rather than tracking volume smoothly.³²

³⁰As an alternative to the time effects I control directly for contemporaneous demand in other fixed income annuity products, $\ln q_t^{FIA}$ (Figure A2, Appendix C), alongside the common maturity effect $\ln Q_t^{DIA}$:

$$\ln S_{a,d,g,i,t} = \gamma_a + \gamma_d + \gamma_i + \gamma_g + \lambda \ln q_t^{FIA} + \beta \ln Q_t + \delta \mathbb{1}_{[a>45]} \times \ln Q_t + u_{a,d,g,i,t}. \quad (22)$$

This delivers similar estimates (Appendix C).

³¹Although annuities are homogeneous, insurers may differ in customer service or integration with other financial services. Insurer effects also absorb time-invariant product differentiation and any time-invariant incentive to price strategically—Section 2.5 returns to this.

³²Including $\log Q$ as a control connects to the role of cumulative demand in Section 2.2.1, although even there past demand has a potentially non-linear effect, since it affects consumer demand non-linearly through signal precision and demand curves are themselves non-linear.

	(1)	(2)	(3)
$\mathbb{1}[a > 45] \ln Q_t$	−0.322 (0.091)	−0.368 (0.105)	−0.380 (0.106)
Wild bootstrap p-value	0.010	0.004	0.004
Observations	5,058	5,058	5,058
R ²	0.420	0.673	0.703
<i>FEs</i>			
Quarter-Year	Yes	Yes	Yes
Gender	No	Yes	Yes
Deferral Age	No	Yes	Yes
Insurer	No	No	Yes

Notes: Standard errors are two-way clustered at the Insurer and Time Period level with p-values constructed using the wild bootstrap (MacKinnon et al. 2021). Estimates reported for coefficient δ in equation (21). Implied selection is described in equation (20) and panel data collected from policy quotes printed in *Annuity Shopper* magazine. All specifications additionally control for purchase age indicators.

TABLE 1. Adverse Selection and Market Maturity

The second column adds gender and deferral-age controls. These raise the R-squared substantially—actuarially fair pricing varies across these characteristics—but have only a modest impact on the differential. It increases slightly yet stays close to the first column, because these characteristics sort annuitants into systematically different pools that nonetheless evolve similarly over time.³³

The final column adds insurer fixed effects, the preferred specification, absorbing compositional change across insurers facing different pools or administrative costs. The estimate increases slightly to $\hat{\delta} = -0.380$, with the across-column differences statistically insignificant. Many are national insurers covering the whole US population and so face similar pools, and the administrative-cost differences that do exist (Mitchell et al. 1999; Kojen and Yogo 2015) are only weakly correlated with entry here. Across all controls the differential is stable and significant at the 1% level. The empirical evidence in this market highlights that adverse selection can be higher among early adopters.

Table 2 estimates equation (23), which splits the differential into separate age-55 and age-65 effects. As with the results in Table 1, the parameter estimates differ slightly

³³Estimated effects are qualitatively similar when estimated separately for each gender and deferral-age grouping, exploiting the same variation in private information across purchase ages, though with less precision in some cases (each subsample is roughly one-tenth of the pooled sample).

	(1)	(2)	(3)
$\mathbb{1}[a = 55] \ln Q_t$	−0.201 (0.076)	−0.246 (0.087)	−0.258 (0.089)
$\mathbb{1}[a = 65] \ln Q_t$	−0.474 (0.124)	−0.523 (0.139)	−0.535 (0.140)
Wild bootstrap p-value, age-55	0.076	0.041	0.037
Wild bootstrap p-value, age-65	0.005	0.004	0.002
Wild bootstrap p-value, 55=65	0.000	0.000	0.000
Observations	5,058	5,058	5,058
R ²	0.432	0.686	0.716
<i>FEs</i>			
Quarter-Year	<i>Yes</i>	<i>Yes</i>	<i>Yes</i>
Gender	<i>No</i>	<i>Yes</i>	<i>Yes</i>
Deferral Age	<i>No</i>	<i>Yes</i>	<i>Yes</i>
Insurer	<i>No</i>	<i>No</i>	<i>Yes</i>

Notes: Standard errors are two-way clustered at the Insurer and Time Period level with p-values constructed using the wild bootstrap (MacKinnon et al. 2021). Estimates reported for coefficient δ_{55} and δ_{65} in equation (23). Implied selection described in equation (20) and panel data collected from policy quotes printed in *Annuity Shopper* magazine. All specifications additionally control for purchase age.

TABLE 2. Adverse Selection and Market Maturity by Private Information

when controls for contract characteristics are added, but are broadly stable.

The age-55 differential ranges from −0.201 to −0.258, and the age-65 differential from −0.474 to −0.535. For all combinations of contract level controls, both are consistently negative. The age-65 differential is significant at the 1% level and the age-55 differential at the 5% level once contract characteristics are included. The decline scales monotonically with purchase age, and hence with private information, with δ_{65} significantly below δ_{55} —jointly testing both predictions and corroborating the identifying assumption that younger groups hold less private information.

4.5. Threats to Identification and Alternative Mechanisms

4.5.1. Horizon Effects

A potential threat to interpreting these results as driven by private-information is that the differentials reflect the deferral horizon of the contract rather than mortality-

relevant private information. At a given deferral age, older purchasers mechanically hold contracts closer to the onset of payouts. If the maturity–selection relationship varies systematically with horizon, e.g., if longer-dated contracts loading more heavily on uncertain long-run mortality, the pooled differential in equation (21) could partly capture horizon rather than private information.

I therefore re-estimate equation (21) separately by the payout horizon, $h = d - a$. This isolates differential effect of maturity across purchase ages within a single homogeneous payout horizon contract. Table 3 reports the results: the first column reproduces the preferred pooled estimate of -0.380 , and the remaining columns estimate the differential for horizons from twenty to thirty years. It is negative and significant for every horizon, ranging from -0.554 to -0.368 . It is largest for the nearer-dated contracts and attenuates at the longest horizon. The combinatorics of the purchase-deferral age pairings mean it is not possible to test horizon-specific differentials at other horizons (Appendix C.2 shows results are not driven by specific deferral ages). Compression of distant cash flows predicts this attenuation, since under any discount rate payouts three decades out carry the least recoverable variation in money’s worth. Holding horizon fixed, the decline cannot be an artifact of differing payout horizons across purchase ages.

	Payout Horizon Specific Estimates			
	(1) Base	(2) h=20	(3) h=25	(4) h=30
$\mathbb{1}[a > 45] \ln Q_t$	-0.380 (0.106)	-0.554 (0.127)	-0.446 (0.140)	-0.368 (0.113)
Wild bootstrap p-value	0.004	0.001	0.005	0.016
Observations	5,058	1,147	767	748
R ²	0.703	0.747	0.690	0.759

Notes: Standard errors are two-way clustered at the Insurer and Time Period level with p-values constructed using the wild bootstrap (MacKinnon et al. 2021). Estimates reported for coefficient δ in equation (21). Implied selection described in equation (20) and panel data collected from policy quotes printed in *Annuity Shopper* magazine. All specifications additionally control for purchase age indicators.

TABLE 3. Constant Horizon Tests for Adverse Selection

4.5.2. Temporal Variation in Selection Measures

Figure 6 plots the period-by-period pricing deviations identifying δ , the age-55 and age-65 gaps in log selection relative to age 45 (Panel A) and their difference (Panel B), against log cumulative considerations.

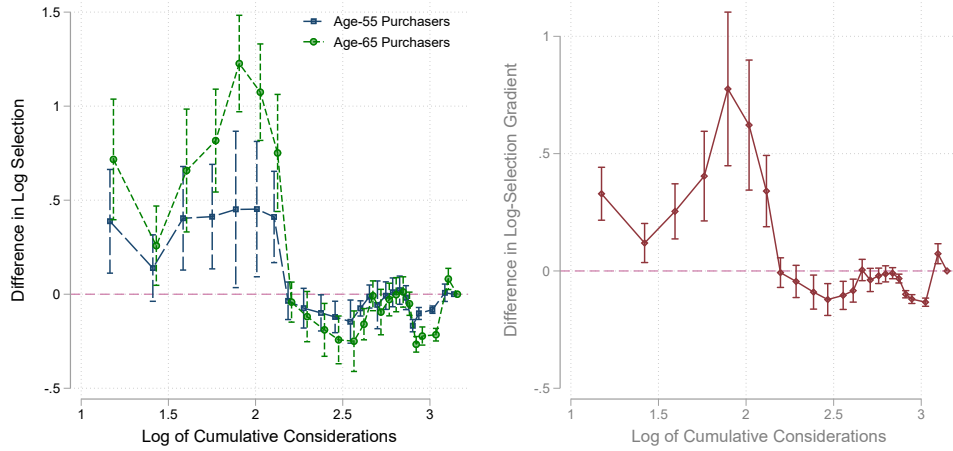
The gaps are large and positive while the market is immature, then collapse to near zero once cumulative sales pass roughly \$8 billion. This decline is concentrated at the transition to a mature market rather than spread smoothly across scale. Because that transition also coincides with insurer entry and the change in broker fiduciary duty, Figure 6 should not be read as isolating the effect of cumulative volume. The time fixed effects in equation (21) absorb the component these shocks share across purchase ages.

Instead, it isolates the *purchase-age ordering* of the decline. In the immature market the age-65 gap is substantially larger than the age-55 gap (Panel A), their difference is positive and distinguishable from zero (Panel B), and both vanish after the transition. A shock common to all purchase ages within a period—the fiduciary rule, entry, or any aggregate factor—is unlikely to produce a differential that scales with purchase age, let alone one that scales in the private-information ranking selection predicts. Two features further favor a selection reading over an intermediation-driven one. The measure is built from insurers’ posted quotes, not realized buyer choices, whereas the fiduciary rule operated on brokers’ steering of buyers—a demand-side channel that does not mechanically enter posted payouts. Second, the rule was vacated in 2018. A steering channel implies the age gradient may re-open after the vacatur, which it does not.

4.5.3. Evidence In Support of the Identifying Assumption

The key identifying assumption, that mortality-relevant private information is larger at older purchase ages, is what separates the dynamics of selection in a maturing market from other time-varying factors. I use subjective mortality expectations from the Health and Retirement Study for a representative sample of potential annuitants aged 50 to 70, capturing information outside the posted price schedule to assess this. Following Hurd and McGarry (2002), I regress a realized 10-year mortality indicator on stated survival beliefs, age, and their interaction, which measures how predictive accuracy varies by age. If older individuals hold more precise private information, realized mortality declines with beliefs (Appendix D provides more details).

Table 4 reports the estimates. Once health and demographic controls enter (columns 2–5), the predictive effect at age 50 becomes insignificant, as health observables absorb



(A) Age 55 and 65 Purchases vs Age 45

(B) Age 65 vs Age 55

Notes: Each point is the period-specific gap in log selection between older and age-45 purchasers, estimated with quarter fixed effects; bands are 95% confidence intervals from standard errors two-way clustered by insurer and time period. Panel A reports the age-55 and age-65 gaps relative to age 45; Panel B their difference. The horizontal axis orders periods by log cumulative considerations and does not enter the regression. Implied selection follows equation (20); *Annuity Shopper* quotes, 2013Q3–2020Q1.

FIGURE 6. Measured Selection By Market Maturity

mortality-relevant information for younger respondents. The interaction, however, remains negative and highly significant throughout—beliefs grow more predictive with age even conditional on observables. At age 65 the implied marginal effect is roughly -0.06 , so a 10 percentage point higher survival belief predicts a 0.6 percentage point lower 10-year mortality. Because annuity pricing conditions only on age and gender, the *full* predictive content of beliefs, not just the residual after controls, is selection-relevant. The HRS data does not allow for testing the identifying assumption at age 45, but evidence on health inequality by age (Hosseini et al. 2022; Russo et al. 2026) suggests extrapolation is innocuous. In addition, the pattern is not driven by focal responses or “flatness” bias (Elder 2013), as I show in Appendix D. That predictive power rises with age validates the identifying assumption.

4.5.4. Robustness

The results are robust to the choice of mortality table and to the opportunity cost of capital, including the Treasury yield the Department of Labor specifies for defined-contribution disclosure (Appendix C). Discounting each horizon at its own zero-coupon risk-free rate attenuates the estimate to zero, but because the is only modest variation

	(1)	(2)	(3)	(4)	(5)
Belief _{<i>i,t</i>}	−0.106*** (0.011)	−0.014 (0.011)	−0.012 (0.011)	−0.012 (0.011)	−0.011 (0.010)
Belief _{<i>i,t</i>} × Age _{<i>i,t</i>}	−0.006*** (0.001)	−0.004*** (0.001)	−0.005*** (0.001)	−0.005*** (0.001)	−0.003*** (0.001)
<i>Controls</i>					
Health	No	Yes	Yes	Yes	Yes
Demographics	No	No	Yes	Yes	Yes
Education	No	No	No	Yes	Yes
Wave FE	No	No	No	No	Yes
N	109,368	109,343	109,316	109,305	109,305
R ²	0.136	0.195	0.204	0.204	0.299

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Data from the Health and Retirement Study, waves 1992–2018 estimating equation A61 with standard errors clustered by individual. Dependent variable is an indicator for death within 10 years. Sample restricted to respondents aged 50–70 for whom mortality outcomes are resolved. Column 1 includes no controls, remaining columns add self-reported health indicators, gender and race controls, education, and wave fixed effects, controlling for systematic mortality improvements.

TABLE 4. Mortality-Relevant Private Information and Age

in long-horizon risk-free rates over the sample this compresses the money’s-worth variation the test exploits rather than correcting a confound—as the within-contract estimates (Table 3) and the discount-rate-free belief evidence (Table 4) confirm.

Alternative Mechanisms. The design rules out a broad class of these by construction: the time fixed effects λ_t absorb any force common across purchase ages, so any confound must work *differentially* by purchase age, in the private-information ranking selection predicts. Appendix C discusses all the concerns below in more detail.

Differential markups. Markup compression from rising entry (Figure 4B) is plausibly *larger* for younger, more elastic buyers, so relative price reductions for older purchasers operate against the finding here. Moreover the differential is unchanged when time effects are replaced with aggregate-demand controls (Appendix C).

Insurer learning about the risk pool would have to operate differentially by age as the market matures—the model’s own prediction, not an alternative—and is implausible given that the deferred-annuity payout formula (20) is nearly identical to those for the annuity and life contracts insurers already price.

Invest-then-harvest strategies predict a *falling* number of insurers, contradicted by the entry in Figure 4. Additionally, undercutting early raises money’s worth while the

market is immature and so lowers measured early selection.

Cross-product subsidization. The market is concentrated among multi-product mutual and large national insurers (New York Life alone reports a 33% share), raising the concern that deferred annuities are priced as retention “add-ons” rather than for standalone profitability. The channel is common across purchase ages. The differential decline among non-mutual and smaller insurers—for whom cross-subsidization is least applicable—is indistinguishable from that among mutual and large-traditional insurers, with the insurer interaction tightly estimated around zero (Appendix C.1.1).

5. Discussion

This paper identifies a novel dynamic free-rider problem in adversely selected markets. When early adopters are more adversely selected, initial entrants bear losses while later competitors wait for the market to mature and bid away the long-run rents that would repay them. Entry can thus unravel even when the mature market would be efficient, leaving markets permanently missing—a dynamic counterpart to [Akerlof \(1970\)](#).

I microfound these entry incentives in a standard insurance model with multidimensional heterogeneity, providing non-parametric sufficient conditions for ordering adversely selected markets by equilibrium prices, quantities, selection, and welfare. These conditions accommodate commonly used empirical specifications and map directly into the graphical framework of [Einav and Finkelstein \(2011\)](#), extending its comparative statics to dynamic and cross-market settings. Together they rationalize low utilization, sluggish take-up, and market inactivity despite substantial unmet demand.

Evidence from the deferred income annuity market is consistent with early adopters being more adversely selected. I exploit age-based differences in mortality-relevant private information, validated using subjective mortality beliefs, to separate the selection component of price changes from secular pricing trends and alternative markup dynamics. The selection component is negative and scales with private information, matching the model’s prediction that selection declines fastest as the market matures where private information is greatest. Although the analysis focuses on insurance, the mechanism applies to any selection market where familiarity with the product grows over time. More broadly, the framework delivers predictions wherever risk premia shift across markets or over time: as [Blundell et al. \(2026\)](#) show in durable goods markets, such shifts compress the lemons penalty, and the comparative statics developed here provide the microfoundations for these related results.

References

- Aghion, P., A. Bergeaud, and J. Van Reenen (2023). The impact of regulation on innovation. *American Economic Review* 113(11), 2894–2936.
- Akerlof, G. A. (1970). The market for "lemons": Quality uncertainty and the market mechanism. *The Quarterly Journal of Economics* 84(3), 488–500.
- Ameriks, J., J. Briggs, A. Caplin, M. D. Shapiro, and C. Tonetti (2016). The long-term-care insurance puzzle: Modeling and measurement. *NBER Working Paper*.
- Arrow, K. J. (1962). Economic welfare and the allocation of resources for invention. In *Readings in industrial economics: Volume two: Private enterprise and state intervention*, pp. 219–236. Springer.
- Attar, A., T. Mariotti, and F. Salanié (2021). Entry-proofness and discriminatory pricing under adverse selection. *American Economic Review* 111(8), 2623–2659.
- Azevedo, E. M. and D. Gottlieb (2017). Perfect competition in markets with adverse selection. *Econometrica* 85(1), 67–105.
- Benhabib, J., J. Perla, and C. Tonetti (2021). Reconciling models of diffusion and innovation: A theory of the productivity distribution and technology frontier. *Econometrica* 89(5), 2261–2301.
- Blundell, R., R. Gu, S. Leth-Petersen, H. Low, and C. Meghir (2026). Durables and lemons: Private information and the market for cars. *Quantitative Economics* 17(1), 38–91.
- Boyer, M. M., P. De Donder, C. Fluett, M.-L. Leroux, and P.-C. Michaud (2020). Long-term care insurance: Information frictions and selection. *American Economic Journal: Economic Policy* 12(3), 134–169.
- Briggs, J., C. Rogers, and C. Tonetti (2023). Risky insurance: Life-cycle insurance portfolio choice with incomplete markets.
- Brown, J. R., A. Kapteyn, E. F. P. Luttmer, O. S. Mitchell, and A. Samek (2021). Behavioral Impediments to Valuing Annuities: Complexity and Choice Bracketing. *The Review of Economics and Statistics* 103(3), 533–546.
- Brown, J. R. and J. M. Poterba (2006). Household ownership of variable annuities. *Tax Policy and the Economy* 20, 163–191.
- Cameron, A. C., J. B. Gelbach, and D. L. Miller (2011). Robust inference with multiway clustering. *Journal of Business & Economic Statistics* 29(2), 238–249.
- Cannon, E. and I. Tonks (2016). Cohort mortality risk or adverse selection in annuity markets? *Journal of Public Economics* 141, 68–81.
- Chade, H. and E. Schlee (2012). Optimal insurance with adverse selection. *Theoretical Economics* 7(3), 571–607.
- Chade, H. and J. Swinkels (2021). Screening in vertical oligopolies. *Econometrica* 89(3), 1265–1311.
- Chetty, R., J. N. Friedman, and E. Saez (2013). Using differences in knowledge across neighborhoods to uncover the impacts of the eitc on earnings. *American Economic Review* 103(7), 2683–2721.
- Chiappori, P.-A. and B. Salanie (2000). Testing for asymmetric information in insurance markets. *Journal of political Economy* 108(1), 56–78.
- Chiappori, P.-A. and B. Salanié (2013). Asymmetric information in insurance markets: Predictions and tests. *Handbook of insurance*, 397–422.

- Cocco, J. F. and P. Lopes (2015). Reverse mortgage design. In *2015 Meeting Papers*, Volume 632.
- Cocco, J. F. and P. Lopes (2020). Aging in place, housing maintenance, and reverse mortgages. *The Review of Economic Studies* 87(4), 1799–1836.
- Cohen, A. and L. Einav (2007). Estimating risk preferences from deductible choice. *American Economic Review* 97(3), 745–788.
- Dorfman, M. S. (1971). The product performance of the life insurance industry: Revisited. *Journal of Risk and Insurance*, 613–626.
- Duflo, E. and E. Saez (2003). The role of information and social interactions in retirement plan decisions: Evidence from a randomized experiment. *The Quarterly Journal of Economics* 118(3), 815–842.
- Egan, M., S. Ge, and J. Tang (2022). Conflicting interests and the effect of fiduciary duty: Evidence from variable annuities. *The Review of Financial Studies* 35(12), 5334–5386.
- Einav, L. and A. Finkelstein (2011). Selection in insurance markets: Theory and empirics in pictures. *Journal of Economic perspectives* 25(1), 115–138.
- Einav, L. and A. Finkelstein (2023). Empirical analyses of selection and welfare in insurance markets: a self-indulgent survey. *The Geneva Risk and Insurance Review* 48(2), 167–191.
- Einav, L., A. Finkelstein, and M. R. Cullen (2010). Estimating welfare in insurance markets using variation in prices. *The Quarterly Journal of Economics* 125(3), 877–921.
- Einav, L., A. Finkelstein, and N. Mahoney (2021). The IO of selection markets. In *Handbook of Industrial Organization*, Volume 5, pp. 389–426. Elsevier.
- Einav, L., A. Finkelstein, and P. Schrimpf (2010). Optimal mandates and the welfare cost of asymmetric information: Evidence from the uk annuity market. *Econometrica* 78(3), 1031–1092.
- Elder, T. E. (2013). The predictive validity of subjective mortality expectations: Evidence from the health and retirement study. *Demography* 50(2), 569–589.
- Fang, H. and Z. Wu (2018). Multidimensional private information, market structure, and insurance markets. *The RAND Journal of Economics* 49(3), 751–787.
- Finkelstein, A. and J. Poterba (2002). Selection effects in the united kingdom individual annuities market. *The Economic Journal* 112(476), 28–50.
- Finkelstein, A. and J. Poterba (2004). Adverse selection in insurance markets: Policyholder evidence from the uk annuity market. *Journal of Political Economy* 112(1), 183–208.
- Geruso, M. and T. J. Layton (2017). Selection in health insurance markets and its policy remedies. *Journal of Economic Perspectives* 31(4), 23–50.
- Geruso, M., T. J. Layton, G. McCormack, and M. Shepard (2023). The two-margin problem in insurance markets. *The Review of Economics and Statistics* 105(2), 237–257.
- Griffith, R. and G. Macartney (2014). Employment protection legislation, multinational firms, and innovation. *Review of Economics and Statistics* 96(1), 135–150.
- Handel, B., I. Hendel, and M. D. Whinston (2015). Equilibria in health exchanges: Adverse selection versus reclassification risk. *Econometrica* 83(4), 1261–1313.
- Handel, B. R. (2013). Adverse selection and inertia in health insurance markets: When nudging hurts. *American Economic Review* 103(7), 2643–2682.
- Handel, B. R., J. T. Kolstad, and J. Spinnewijn (2019). Information frictions and adverse selection:

- Policy interventions in health insurance markets. *Review of Economics and Statistics* 101(2), 326–340.
- Hendren, N. (2013). Private information and insurance rejections. *Econometrica* 81(5), 1713–1762.
- Hosseini, R., K. A. Kopecky, and K. Zhao (2022). The evolution of health over the life cycle. *Review of Economic Dynamics* 45, 237–263.
- Hurd, M. D. and K. McGarry (2002). The predictive validity of subjective probabilities of survival. *The Economic Journal* 112(482), 966–985.
- Israel, M. (2005). Services as experience goods: An empirical examination of consumer learning in automobile insurance. *American Economic Review* 95(5), 1444–1463.
- Koijen, R. S. and M. Yogo (2015). The cost of financial frictions for life insurers. *American Economic Review* 105(1), 445–475.
- Kong, E., T. Layton, and M. Shepard (2024). Adverse selection and (un) natural monopoly in insurance markets. Technical report, National Bureau of Economic Research.
- Leive, A., L. Friedberg, and B. Davis (2022). Overpaying and undersaving? correlated mistakes in retirement saving and health insurance choices. Technical Report 2022-06, Wharton Pension Research Council Working Paper.
- Lockwood, L. M. (2012). Bequest motives and the annuity puzzle. *Review of Economic Dynamics* 15(2), 226–243.
- Lockwood, L. M. (2018). Incidental bequests and the choice to self-insure late-life risks. *American Economic Review* 108(9), 2513–2550.
- MacKinnon, J. G., M. Ø. Nielsen, and M. D. Webb (2021). Wild bootstrap and asymptotic inference with multiway clustering. *Journal of Business & Economic Statistics* 39(2), 505–519.
- Mahoney, N. and E. G. Weyl (2017). Imperfect competition in selection markets. *Review of Economics and Statistics* 99(4), 637–651.
- McGuire, T. G., S. Schillo, and R. C. van Kleef (2020). Reinsurance, repayments, and risk adjustment in individual health insurance: Germany, the Netherlands, and the U.S. Marketplaces. *American Journal of Health Economics* 6(1), 139–168.
- Michaud, P.-C. and P. S. Amour (2023). Longevity, health and housing risks management in retirement. Technical report.
- Milgrom, P. R. (1981). Good news and bad news: Representation theorems and applications. *The Bell Journal of Economics*, 380–391.
- Milgrom, P. R. and R. J. Weber (1982). A theory of auctions and competitive bidding. *Econometrica*, 1089–1122.
- Mitchell, O. S., J. M. Poterba, M. J. Warshawsky, and J. R. Brown (1999). New evidence on the money's worth of individual annuities. *American Economic Review* 89(5), 1299–1318.
- Murray, M. L. (1976). The theory and practice of innovation in the private insurance industry. *Journal of Risk and Insurance*, 653–671.
- Nakajima, M. and I. A. Telyukova (2017). Reverse mortgage loans: A quantitative analysis. *The Journal of Finance* 72(2), 911–950.
- Nelson, R. R. (1959). The simple economics of basic scientific research. *Journal of political economy* 67(3), 297–306.
- O'Dea, C. and D. Sturrock (2023). Survival pessimism and the demand for annuities. *The Review*

- of *Economics and Statistics* 105(2), 442–457.
- Pashchenko, S. (2013). Accounting for non-annuitization. *Journal of Public Economics* 98, 53–67.
- Poterba, J. M. and A. Solomon (2021). Discount rates, mortality projections, and money’s worth calculations for us individual annuities. *NBER Working Paper*.
- Roodman, D., M. Ø. Nielsen, J. G. MacKinnon, and M. D. Webb (2019). Fast and wild: Bootstrap inference in stata using boottest. *The Stata Journal* 19(1), 4–60.
- Rothschild, M. and J. Stiglitz (1976). Equilibrium in competitive insurance markets: An essay on the economics of imperfect information. *The Quarterly Journal of Economics* 90(4), 629–649.
- Rudelius, W. and G. L. Wood (1970). Life insurance product innovations. *Journal of Risk and Insurance*, 169–184.
- Russo, N., R. McGee, M. De Nardi, M. Borella, and R. Abram (2026). Midlife health and later life economic inequality. *NBER Working Paper*.
- Solomon, A. (2023). Imperfect private information in insurance markets. *Review of Economics and Statistics*, 1–46.
- Spinnewijn, J. (2017). Heterogeneity, demand for insurance, and adverse selection. *American Economic Journal: Economic Policy* 9(1), 308–343.
- Starc, A. (2014). Insurer pricing and consumer welfare: Evidence from medigap. *The RAND Journal of Economics* 45(1), 198–220.
- Tchen, A. H. (1980). Inequalities for distributions with given marginals. *The Annals of Probability*, 814–827.
- Thompson, J. R. (2010). Counterparty risk in financial contracts: Should the insured worry about the insurer? *The Quarterly Journal of Economics* 125(3), 1195–1252.
- van de Ven, W. P. M. M. and F. T. Schut (2011). Guaranteed access to affordable coverage in individual health insurance markets. In S. Glied and P. Smith (Eds.), *Oxford Handbook of Health Economics*, Chapter 17. Oxford University Press.
- Veiga, A. and E. G. Weyl (2016). Product design in selection markets. *The Quarterly Journal of Economics* 131(2), 1007–1056.
- Warshawsky, M. (1988). Private annuity markets in the united states: 1919-1984. *Journal of Risk and Insurance*, 518–528.
- Webb, A. (2011). The United States Longevity Insurance Market. In *Securing Lifelong Retirement Income: Global Annuity Markets and Policy*. Oxford University Press.
- Williams, H. L. (2017). How do patents affect research investments? *Annual Review of Economics* 9, 441–469.
- Zinman, J. (2014). Consumer credit: Too much or too little (or just right)? *The Journal of Legal Studies* 43(S2), S209–S237.

Online Appendix

Appendix A. Proofs for Section 2

This section provides general versions of the results established under normality in Section 2.3 and extends them to non-parametric families of joint distributions.

A.1. Proofs for Need-Based Early Adoption in Section 2.2.1

PROOF OF PROPOSITION 1. By the definition of the copula, strictly increasing transformation u and the location $\mu(S)$ leave the copula of $(\pi_i, \omega_i(S))$ unchanged. Therefore, it is the copula of the index $g(\pi_i) + \hat{v}_i(S)$ paired with π_i .

Since $\hat{v}_i(S) \perp \pi_i$ and

$$\text{Var}(\hat{v}_i(S)) = \frac{\sigma_v^4}{\sigma_v^2 + \sigma_\zeta^2(S)}$$

is strictly increasing in S —rising from 0 as $\sigma_\zeta^2(S) \rightarrow \infty$ to σ_v^2 as $\sigma_\zeta^2(S) \rightarrow 0$ —the copula depends on S only through this variance. Under joint normality the index is jointly Gaussian with π_i , with correlation

$$\rho(S) = \frac{g' \sigma_\pi}{\sqrt{g'^2 \sigma_\pi^2 + \text{Var}(\hat{v}(S))}},$$

which is strictly decreasing in S . The Gaussian copula is strictly increasing in ρ pointwise ($\partial_\rho \Phi_\rho = \Phi_\rho > 0$), so C^S is strictly decreasing in S in the concordance order. Hence $C^{IM}(u, v) \geq C^{MM}(u, v)$ for all (u, v) , which is Assumption 3 on the relevant price domain under adverse selection. $\square$

LEMMA A1 (Location Channel and Demand Expansion). *Maintain the conditions of Proposition 1. The location channel $\mu(S)$, with $d\mu(S)/ds > 0$, shifts the willingness-to-pay distribution rightward without affecting the copula C^S . The immature and mature willingness-to-pay distributions are ordered by a mean–variance shift and cross exactly once, at*

$$p^c = \frac{\sigma^{MM} m^{IM} - \sigma^{IM} m^{MM}}{\sigma^{MM} - \sigma^{IM}},$$

where $m(S) = E[g(\pi_i)] + \mu(S)$ and $\sigma^2(S) = \text{Var}(g(\pi_i)) + \text{Var}(\hat{v}_i(S))$. On the price domain $[\underline{p}, \infty)$, bounded below by the lowest attainable average cost $\underline{p} \equiv \mu_\pi$ and hence containing every

candidate equilibrium price, $p^c \leq \underline{p}$ implies $D^{MM}(p) \geq D^{IM}(p)$ for all $p \geq \underline{p}$, which is Assumption 2.

PROOF OF LEMMA A1. The transform u and the location $\mu(S)$ leave C^S unchanged, as in the proof of Proposition 1, so $\mu(S)$ acts only on the marginal of ω . The willingness-to-pay index is normally distributed with mean $m(S) = E[g(\pi_i)] + \mu(S)$ and variance $\sigma^2(S) = \text{Var}(g(\pi_i)) + \text{Var}(\hat{v}_i(S))$.

Since $d\mu(S)/dS > 0$ and $\text{Var}(\hat{v}_i(S))$ is increasing in S , $m^{MM} > m^{IM}$ and $\sigma^{MM} > \sigma^{IM}$. Because Φ is strictly increasing, $F_\omega^{MM}(p) \leq F_\omega^{IM}(p)$ if and only if $(p - m^{MM})/\sigma^{MM} \leq (p - m^{IM})/\sigma^{IM}$, a linear inequality in p that holds if and only if $p \geq p^c$.

The two CDFs therefore cross exactly once, at p^c , with $F_\omega^{MM}(p) \leq F_\omega^{IM}(p)$ —equivalently $D^{MM}(p) \geq D^{IM}(p)$ —for all $p \geq p^c$. Under average-cost pricing every candidate equilibrium price satisfies $p = AC(p) \geq \mu_\pi = \underline{p}$, so $p^c \leq \underline{p}$ delivers demand expansion at every such price. $\square$

Two observations are worth discussing. First, the restriction $p^c \leq \underline{p}$ removes demand reversal, driven by the Gaussian tail, from the equilibrium-relevant domain entirely: average-cost pricing confines every candidate equilibrium price to $[\mu_\pi, \infty)$, and below $p^c \leq \mu_\pi$ the price sits beneath the lowest break-even average cost, where take-up is high in both markets and the level comparison cannot bear on any equilibrium. Second, the single crossing is an artifact of the Gaussian signal-extraction device rather than the central mechanism. The substantive content of the microfoundation is the concordance ordering of Proposition 1, a statement about dependence that is exact, global, and invariant to the marginals, to u , and to $\mu(S)$; the level comparison inherits only the limitation common to every normal-distribution result in this paper, that unbounded support forces either a fixed dispersion (Assumption 4) or a bounded price domain, and is available with truncated tails (Levy 1982). The general copula theory of the following subsections assumes Assumptions 2–3 directly and dispenses with the Gaussian entirely, so the tail behavior of this special case has no substantive implication for the general comparative statics.

A.2. General Proofs and Additional Results for Sections 2.2–2.5: Within Markets

The Role of the MLRP assumption. The MLRP property is widely invoked in studies of incomplete information and has a convenient intuitive interpretation. The proof of Proposition 4 establishes that monotonicity of the cost curves are each available under

milder assumptions on the copula that are directly implied by the stronger MLRP.¹

PROOF OF PROPOSITION 4. Preliminaries. Let copula C denote the dependence between cost type, π_i , and willingness-to-pay, ω_i . Then the monotone likelihood ratio dependence implies (see, e.g., [Nelsen 2006](#) Theorem 5.2.19):

- i. [Component-wise Concavity] $C(u, v)$ is concave in v for all $u \in (0, 1)$.
- ii. [Corner Set Monotonicity] $(u - C(u, 1 - v))/v$ is increasing in v for all $u \in (0, 1)$.

Proof of Part (i). Beginning with the monotonicity of marginal costs – the CDF of the conditional random variable $\pi|\omega = p$ is

$$\begin{aligned} F_{\pi|\omega}(\pi|\omega) &\equiv \int_{-\infty}^{\pi} f_{\pi|\omega}(u|\omega) du = \int_{-\infty}^{\pi} \frac{f_{\pi,\omega}(u, \omega)}{f_{\omega}(\omega)} du \\ &= C_2(F_{\pi}(\pi), F_{\omega}(\omega)) = g(F_{\pi}(\pi)), \end{aligned} \quad (\text{A1})$$

for a given copula C where $C_2(x, y) = \partial C(x, y)/\partial y$.

As the marginal cost curve is a conditional expectation $MC(p) = E(\pi|\omega = p)$, stochastic dominance of the conditional distributions is a sufficient condition to order marginal costs in price

$$F_{\pi|\omega}(\pi|\omega = v_1) \succeq F_{\pi|\omega}(\pi|\omega = v_2) \quad \forall v_1 \geq v_2 \quad (\text{A2})$$

$$\longrightarrow MC(v_1) = E(\pi|\omega = v_1) \geq E(\pi|\omega = v_2) = MC(v_2) \quad \forall v_1 \geq v_2. \quad (\text{A3})$$

Substituting equation (A1) gives the following necessary and sufficient condition:

$$C_2(u, F_{\omega}(v_1)) \leq C_2(u, F_{\omega}(v_2)) \quad \forall v_1 \geq v_2 \text{ and } \forall u \quad (\text{A4})$$

$$\longleftrightarrow C_2(u, a) \leq C_2(u, b) \quad \forall a \geq b \text{ and } \forall u \quad (\text{A5})$$

as $v_1 \geq v_2 \longrightarrow F_{\omega}(v_1) \geq F_{\omega}(v_2)$. This is satisfied for all copulas where $C_2(\cdot, \cdot)$ is decreasing in its second argument. Condition (i) of Proposition (4) guarantees C is concave and satisfies this property.

This establishes that $MC(p)$ is increasing in p .

¹A weaker necessary condition for the second property is a log-concave survival function – a common assumption when studying incomplete information (for instance [Milgrom and Weber 1982](#), [Chade and Schlee 2012](#) and [Chade and Swinkels 2021](#)).

Proof of Parts (ii) and (iii). By the law of iterated expectations, the average cost curve admits the representation

$$AC(p) = E(\pi | \omega \geq p) = E(MC(\omega) | \omega \geq p). \quad (A6)$$

Since $MC(\cdot)$ is weakly increasing (Part (i)), raising the truncation threshold from v_2 to $v_1 \geq v_2$ shifts the conditional distribution of $\omega | \omega \geq p$ in the sense of first-order stochastic dominance, so

$$AC(v_1) = E(MC(\omega) | \omega \geq v_1) \geq E(MC(\omega) | \omega \geq v_2) = AC(v_2) \quad \forall v_1 \geq v_2, \quad (A7)$$

establishing Part (ii). Moreover, on the event $\{\omega \geq p\}$, monotonicity implies $MC(\omega) \geq MC(p)$ pointwise, so

$$AC(p) = E(MC(\omega) | \omega \geq p) \geq MC(p), \quad (A8)$$

establishing Part (iii). $\square$

COROLLARY A1 (Corner-Set Copula Condition for Average Costs). *The monotonicity of $AC(p)$ in Part (ii) of Proposition 4 can alternatively be established directly from a stochastic dominance ordering. The CDF of the conditional random variable $\pi | \omega > p$ is*

$$F_{\pi | \omega > p}(\pi | \omega > p) \equiv \frac{u - C(u, F_\omega(p))}{1 - F_\omega(p)}. \quad (A9)$$

Substituting equation (A9), this holds whenever the copula satisfies

$$\frac{u - C(u, F_\omega(v_1))}{1 - F_\omega(v_1)} \leq \frac{u - C(u, F_\omega(v_2))}{1 - F_\omega(v_2)} \quad \forall v_1 \geq v_2 \text{ and } \forall u, \quad (A10)$$

which is condition (ii) of Proposition 4. This copula restriction is the relevant sufficient primitive for within-market comparisons of average costs in Proposition A2.

A.3. Additional Results and Examples of Copula

This section shows how to derive the distribution of willingness-to-pay directly from assumptions on cost and risk premia in the linear case. This will depend on the dependence between cost and risk premia, as encoded by the copula $\tilde{C}$, and marginal distributions. The following results are derived in [Cherubini et al. \(2011\)](#) who also provide extensions to multivariate settings (see also [Navarro and Sarabia 2022](#)). These multivariate extensions can also be used to derive expressions for the joint-density of net premia from the underlying densities of risk premia r and choice frictions ε . This extends general results on the convolution of probability distributions.

The distribution function for willingness-to-pay, ω , is given by F_ω

$$F_\omega(\omega) = \int_{-\infty}^{\infty} f_\pi(c) \partial_1 \tilde{C}(F_\pi(c), F_{\tilde{r}}(\omega - c)) dc \quad (\text{A11})$$

provided that $\lim_{v \rightarrow 0^+} \partial_1 \tilde{C}(u, v) = 0$ for all $u \in (0, 1)$. This is a mixture over the conditional distribution of $(\tilde{r} + c | \pi = c)$, which is given by²

$$F_{\omega|\pi}(\omega | \pi = c) = \partial_1 \tilde{C}(F_\pi(c), F_{\tilde{r}}(\omega - c)). \quad (\text{A12})$$

We can use this to derive the following joint CDF

$$F_{\pi, \omega}(t, v) = \int_{-\infty}^t F_{\omega|\pi}(v | s) dF_\pi(s) = \int_{-\infty}^t \partial_1 \tilde{C}(F_\pi(s), F_{\tilde{r}}(v - s)) dF_\pi(s), \quad (\text{A13})$$

which in turn can be used recover C , the copula specifying the dependence between ω and π , by application of Sklar's Theorem:

$$C(u, v) = F_{\pi, \omega} \left(F_\pi^{-1}(u), F_\omega^{-1}(v) \right) = \int_0^u \partial_1 \tilde{C} \left(x, F_{\tilde{r}} \left(F_\omega^{-1}(v) - F_\pi^{-1}(x) \right) \right) dx. \quad (\text{A14})$$

There are not closed form results for this integral as it depends on the marginal densities and the copula. However, we can additionally derive the following probability density functions which will prove useful in evaluating its properties:

$$f_{\omega|\pi}(\omega | \pi = c) = f_{\tilde{r}}(\omega - c) \partial_{1,2} \tilde{C}(F_\pi(c), F_{\tilde{r}}(\omega - c)) \quad (\text{A15})$$

$$f_\omega(\omega) = \int_{-\infty}^{\infty} f_\pi(c) f_{\tilde{r}}(\omega - c) \partial_{1,2} \tilde{C}(F_\pi(c), F_{\tilde{r}}(\omega - c)) dc \quad (\text{A16})$$

²Theorem 2.3 [Navarro and Sarabia 2022](#) shows that this admits a distortion function representation.

and the following corollary uses these representations to re-express Proposition 4 in terms of the joint distributions of π and $\tilde{r}$.

COROLLARY A2 (MLRP). *The monotone-likelihood ratio property for willingness-to-pay, ω , and cost type, π , is equivalent to the following property for the cost types and risk premia $\tilde{r}$:*

$$\frac{f_{\tilde{r}}(\omega - \pi) \partial_{1,2} \tilde{C}(F_{\pi}(\pi), F_{\tilde{r}}(\omega - \pi))}{f_{\tilde{r}}(\omega - \pi') \partial_{1,2} \tilde{C}(F_{\pi}(\pi'), F_{\tilde{r}}(\omega - \pi'))} \geq \frac{f_{\tilde{r}}(\omega' - \pi) \partial_{1,2} \tilde{C}(F_{\pi}(\pi), F_{\tilde{r}}(\omega' - \pi))}{f_{\tilde{r}}(\omega' - \pi') \partial_{1,2} \tilde{C}(F_{\pi}(\pi'), F_{\tilde{r}}(\omega' - \pi'))} \quad \forall \pi' \geq \pi \text{ and } \omega' \geq \omega$$

Thus, all joint distributions satisfying this property inherit properties (i) - (iii) in Proposition 4.

COROLLARY A3 (Primitive Sufficient Condition for the Additive Index). *Suppose $\omega = \pi + \tilde{r}$ and let $h_{\pi}(r) \equiv f_{\tilde{r}|\pi}(r|\pi)$ denote the conditional density of $\tilde{r}$ given π . If for every $\pi' > \pi$:*

i. the family $\tilde{r}|\pi$ satisfies the monotone likelihood ratio property in $(\tilde{r}, \pi)$, i.e.

$$\frac{h_{\pi'}(r)}{h_{\pi}(r)} \text{ is weakly increasing in } r; \quad (\text{A17})$$

ii. for each π , the density $h_{\pi}(\cdot)$ is log-concave.

Then the conditional density $f_{\omega|\pi}(\omega|\pi)$ satisfies the monotone likelihood ratio property in (ω, π) . Hence Proposition 4 holds for ω , and therefore also for any strictly increasing transformation $\omega = g(\omega)$.

PROOF. Write $\Delta = \pi' - \pi > 0$ and $r = \omega - \pi$. Since $f_{\omega|\pi}(\omega|\pi) = h_{\pi}(\omega - \pi)$, the likelihood ratio can be decomposed as

$$\frac{f_{\omega|\pi}(\omega|\pi + \Delta)}{f_{\omega|\pi}(\omega|\pi)} = \underbrace{\frac{h_{\pi+\Delta}(r - \Delta)}{h_{\pi+\Delta}(r)}}_{\text{increasing in } r \text{ by (ii)}} \cdot \underbrace{\frac{h_{\pi+\Delta}(r)}{h_{\pi}(r)}}_{\text{increasing in } r \text{ by (i)}}. \quad (\text{A18})$$

The second term is increasing in r by assumption (i). The first term is increasing in r by log-concavity of $h_{\pi+\Delta}$, since log-concavity implies $\log h_{\pi+\Delta}(r - \Delta) - \log h_{\pi+\Delta}(r)$ is increasing in r .³ Therefore the full likelihood ratio is increasing in ω , establishing MLRP for $\omega|\pi$. The stated conclusions then follow from the MLRP Corollary above. $\square$

This result decouples the restriction from the joint law of (π, ω) to a statement about the conditional family $\tilde{r}|\pi$, which is more directly interpretable in terms of model primitives. Condition (i) requires that higher cost types draw risk premia from

³This is a standard result that log-concavity of a density implies monotonicity of its hazard rate; see, e.g., Milgrom 1981.

stochastically higher distributions in the likelihood ratio sense; condition (ii) is a regularity condition on the shape of each conditional density that is satisfied by all log-concave families, including the normal, exponential, uniform, and logistic.

I now turn to discussing examples of common parametric copulas and the conditional distributions of ranks (summarized by their derivatives). While this allows for convenient theoretical analysis, it is worth emphasizing, however, that in empirical settings copulas can be estimated directly from the data using non-parametric tools. This is an inexhaustive list and other copula family also abound.^{4,5}

Example 1: Gaussian copula. The Gaussian copula is a family of copulas that vary in their correlation parameter ρ . All multivariate Normal distributions have Gaussian copulas and Normal marginals. However, not all normally distributed marginals have Gaussian copulas. The bivariate copula is defined as:

$$C(u, v; \rho) = \int_{-\infty}^{\Phi^{-1}(u)} \int_{-\infty}^{\Phi^{-1}(v)} \frac{1}{2\pi\sqrt{1-\rho^2}} \exp \left[-\frac{s^2 - 2\rho st + t^2}{2(1-\rho^2)} \right] ds dt \quad (\text{A19})$$

$$= \int_{-\infty}^{\Phi^{-1}(u)} \frac{1}{\sqrt{2\pi}} \exp \left[-\frac{s^2}{2} \right] \Phi \left[\frac{\Phi^{-1}(v) - \rho s}{\sqrt{1-\rho^2}} \right] ds \quad (\text{A20})$$

with first derivative given by (it is symmetric)

$$\frac{d}{du} C(u, v; \rho) = \frac{1}{\sqrt{2\pi}} \exp \left[-\frac{\Phi^{-1}(u)^2}{2} \right] \cdot \frac{d \Phi^{-1}(u)}{du} \cdot \Phi \left[\frac{\Phi^{-1}(v) - \rho \Phi^{-1}(u)}{\sqrt{1-\rho^2}} \right] \quad (\text{A21})$$

$$= \Phi \left[\frac{\Phi^{-1}(v) - \rho \Phi^{-1}(u)}{\sqrt{1-\rho^2}} \right] \quad (\text{A22})$$

and cross-partial derivative (or copula density)

$$\frac{\partial^2}{\partial u \partial v} C(u, v; \rho) = \frac{1}{\sqrt{1-\rho^2}} \frac{d}{dv} \Phi^{-1}(v) \Phi' \left[\frac{\Phi^{-1}(v) - \rho \Phi^{-1}(u)}{\sqrt{1-\rho^2}} \right] \quad (\text{A23})$$

⁴Moreover, the convolution of copulas above is closed under mixtures which, with sufficient mixing, allows for almost arbitrary dependence using parametric families.

⁵A less common copula in statistical applications, but highly relevant in economic applications is the Burr copula. This is characterized by joint-Burr distributions which exhibit similar jointness properties to the normal distribution. Moreover, univariate Burr distributions have extensive use as a generalized version of the Pareto distribution to model income distributions with long tails (see, for instance, [Singh and Maddala 1976](#)).

$$= \frac{1}{\sqrt{1-\rho^2}} \exp \left[\frac{-\rho^2 \Phi^{-1}(v)^2 + 2\rho \Phi^{-1}(u) \Phi^{-1}(v) - \rho^2 \Phi^{-1}(u)^2}{2(1-\rho^2)} \right]. \quad (\text{A24})$$

Substituting into Equation A12 gives the following expression for the conditional CDF in this example:

$$F_{\omega|\pi}(\omega|\pi=c) = \Phi \left[\frac{\Phi^{-1}(F_{\tilde{r}}(\omega-c)) - \rho \Phi^{-1}(F_{\pi}(c))}{\sqrt{1-\rho^2}} \right], \quad (\text{A25})$$

which highlights that the normality of the conditional density is a special property. It requires both the specific Gaussian copula and the normally distributed marginals.

Figure A1 shows how the parametric specification of the copula effects the joint distribution when the marginal distribution are fixed. The left columns shows how the joint distribution of π and $\tilde{r}$ changes with the correlation parameter. In both cases I assume π and $\tilde{r}$ are log normally distributed with their logs following standard normals. Comparing panels A and C shows the role of the correlation parameter for the gaussian copula. As the correlation increases, it stretches the density towards the upper right corner.

Example 2: Gumbel Copula. The Gumbel copula has heavier tails than the normal distribution and is also asymmetric, with more weight in the right tail. It's left tail, however, is similar to the normal distribution. The bivariate copula is defined (for $\delta \geq 1$) as:

$$C(u, v; \delta) = \exp \left(- \left[(-\log u)^\delta + (-\log v)^\delta \right]^{1/\delta} \right) \quad (\text{A26})$$

$$= \exp \left(-A^{1/\delta} \right) \quad \text{where} \quad A = (-\log u)^\delta + (-\log v)^\delta \quad (\text{A27})$$

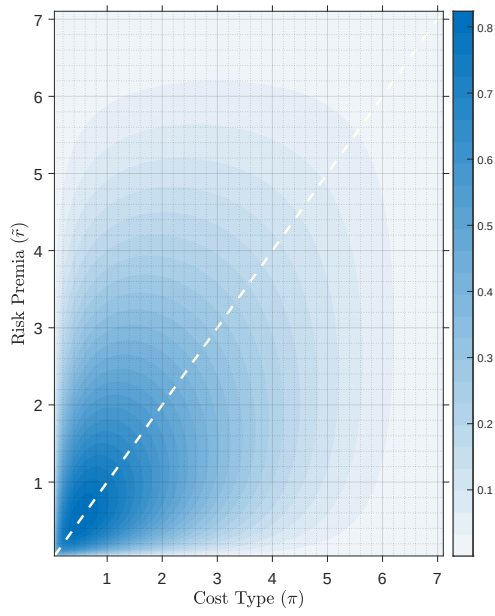
with first derivative given by (it is symmetric)

$$\frac{\partial}{\partial u} C(u, v; \delta) = \exp \left(-A^{1/\delta} \right) \cdot \frac{(1 + (-\log v / -\log u)^\delta)^{1/\delta-1}}{u} \quad (\text{A28})$$

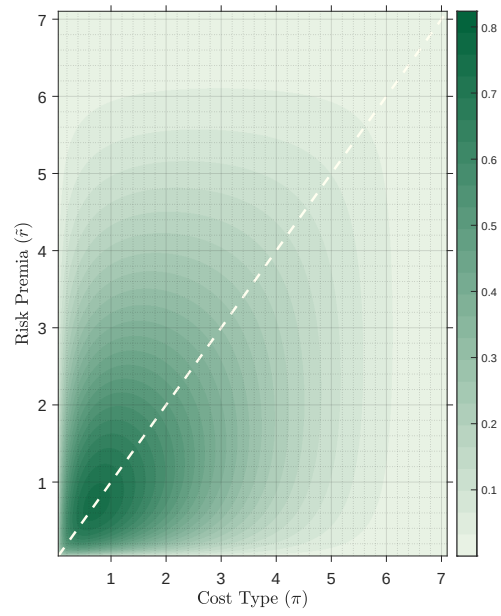
and, letting $A = (-\log u)^\delta + (-\log v)^\delta$, the cross-partial derivative is

$$\frac{\partial^2}{\partial u \partial v} C(u, v; \delta) = C(u, v; \delta) \frac{(-\log u)^{\delta-1} (-\log v)^{\delta-1}}{u v} A^{1/\delta-2} \left(A^{1/\delta} + \delta - 1 \right). \quad (\text{A29})$$

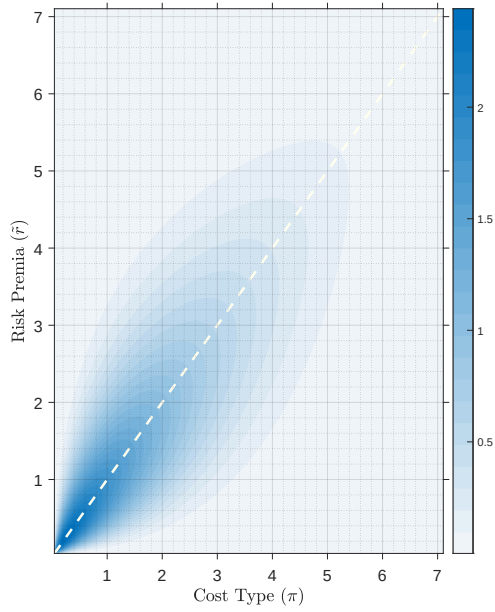
Comparing the right column of Figure A1, which shows the Gumbel copula, to



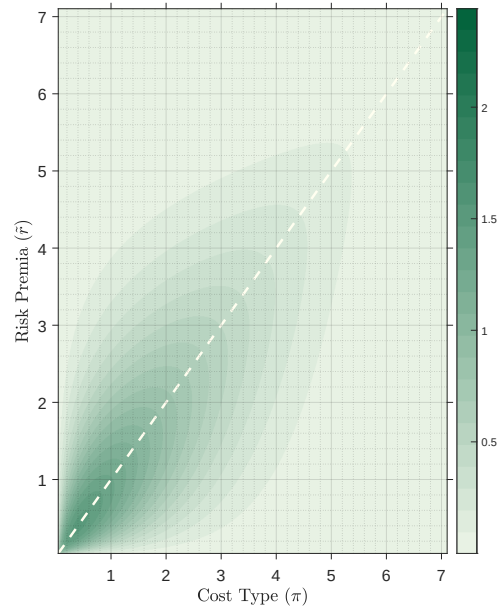
(A) Gaussian Copula (correlation=0.5)



(B) Gumbel Copula (correlation=0.5)



(C) Gaussian Copula (correlation=0.9)



(D) Gumbel Copula (correlation=0.9)

FIGURE A1. Copula and Joint Densities By Pearson Correlation

Notes: Panels vary the copula (and parameter controlling correlation) for the rank dependency between cost type (π) and risk premia ($\tilde{r}$). In all panels the marginal distributions of cost type (π) and risk premia ($\tilde{r}$) are distributed log-normally with a mean of 0 and unit variance.

the left highlights differences induced in the joint density by choice of the copula. At lower correlations, panels A and B, the two copulas deliver more similar joint-densities. However, panel B, using the Gumbel copula, reveals heavier tails of the joint distribution. This is even more pronounced for higher correlations shown in panels C and D. Here, like the Gaussian copula, panel D stretches the density towards the upper-right corner of the figure. In contrast though, the normal exhibits softer contours. Instead, the surface of the joint density is pulled “tauter” under the Gumbel copula and more mass is concentrated in the joint tail. Thus the risk is more jointly heavily tailed.

Example 3: The Independent Copula. The independent copula describes the case where the two random variables are independent. The relevant formulae are:

$$C(u, v) = uv \tag{A30}$$

$$\frac{\partial}{\partial u} C(u, v) = v \tag{A31}$$

$$\frac{\partial^2}{\partial u \partial v} C(u, v) = 1. \tag{A32}$$

A.4. General Proofs and Additional Results for Sections 2.2–2.5: Across Markets

The Role of the Constant dispersion assumption. Constant dispersion in willingness-to-pay is required for the normal distribution as the support is unbounded. Equivalent results are available with truncated tails (Levy 1982). It is not required for the general results under arbitrary marginals.

A.4.1. Demand Contraction: Additional Results

Demand contractions are immediate from assumptions on willingness-to-pay. Instead, I begin by proving demand contraction under the following alternative assumptions on the sub components of willingness-to-pay:

ASSUMPTION A1. *The perceived risk premia (net of any behavioral influences) of insurance contracts are lower for immature markets. $\tilde{r}^{MM} \succeq \tilde{r}^{IM}$*

ASSUMPTION A2. *Risk types, π_i , have identical marginal distributions in mature and immature markets.*

and

ASSUMPTION A3. *Additionally, assume that resorting satisfies the following conditions for the copulas $\tilde{C}$, the bivariate copula denoting the dependence between π_i and $\tilde{r}_i$*

i. *The partial derivatives of $\tilde{C}$ satisfy $\frac{\partial \tilde{C}^{MM}(u, F_{\tilde{r}}^{MM}(\tilde{r}))}{\partial u} \leq \frac{\partial \tilde{C}^{IM}(u, F_{\tilde{r}}^{IM}(\tilde{r}))}{\partial u} \quad \forall u \in [0, 1] \text{ and } \tilde{r}.$*

PROPOSITION A1 (Demand Contraction). *If $\omega_i = u(\pi_i + \tilde{r}_i)$, willingness-to-pay is on average larger and demand expands in mature markets. $D^{IM}(p) \leq D^{MM}(p) \quad \forall p$*

PROOF OF PROPOSITION A1. First consider linear utility with $\omega_i = \pi_i^{IM} + \tilde{r}_i^{IM}$. Assumptions A1 - A3 guarantee that the conditions in Theorem 3.3 in Navarro and Sarabia (2022) are satisfied. A direct application gives

$$\omega_i^{IM} = \pi_i^{IM} + r_i^{IM} + \varepsilon_i^{IM} \preceq \pi_i^{MM} + r_i^{MM} + \varepsilon_i^{MM} = \omega_i^{MM} \longrightarrow F_{\omega}^{MM}(p) \leq F_{\omega}^{IM}(p) \quad \forall p. \quad (\text{A33})$$

As the usual stochastic order is preserved under increasing transformations, the stated proposition then follows immediately for any increasing $u(\cdot)$ $\square$

This level effect does not change the identity of marginal (and inframarginal) consumers at any quantity demanded is the same and the equilibrium price increases

and demand falls (See Proposition 2). Similar arguments apply to assumptions on r and ε .

Moreover, when there is *only* a level effect the following lemma shows that the utility function can be generalized to allow for arbitrary substitution patterns between π_i .

LEMMA A2 (Arbitrary Substitutes). *When there is no re-sorting ($C^{IM} = C^{MM}$), Proposition A1 can be extended to any willingness-to-pay satisfying*

$$\omega_i = u(\pi_i, \tilde{r}_i) \quad (\text{A34})$$

where $u(\cdot, \cdot)$ is a function increasing in its arguments.

PROOF OF LEMMA A2. Identical in distribution satisfies $\pi^{MM} \succeq \pi^{IM}$. Thus, each element of the vector $(\pi_i, \tilde{r}_i)$ in mature markets is larger in the usual stochastic order than in immature markets. By the common copula, it follows that the vector $\mathbf{X}^{IM} \preceq \mathbf{X}^{MM}$ in the usual *multivariate* stochastic ordering (Shaked and Shanthikumar 2007, Theorem 6.B.14).

This order is preserved for any increasing function $u : \mathcal{R}^2 \rightarrow \mathcal{R}$, then $u(\mathbf{X}^{IM}) \preceq u(\mathbf{X}^{MM})$. Thus, $F_\omega^{IM}(p) \geq F_\omega^{MM}(p) \forall p$ and the stated lemma follows. This result extends to any multi-dimensional vector (for example r and ε) under corresponding assumptions on their marginals. $\square$

For the special case of the normal distribution, demand contractions are tightly linked to the price elasticity of consumers in immature and mature markets. I denote this $\tilde{\eta}$ to distinguish from η which denotes the semi-elasticity in the main text. The following corollary formalises this for the general case:

COROLLARY A4 (Price Elasticity of Demand). *When willingness-to-pay is larger in the hazard rate order in mature markets, that is*

$$\frac{f^{IM}(\omega)}{1 - F^{IM}(\omega)} \geq \frac{f^{MM}(\omega)}{1 - F^{MM}(\omega)} \forall p, \quad (\text{A35})$$

then demand is more price sensitive in immature markets.

PROOF. This follows from substituting the definition of demand into the usual formula for the elasticity,

$$\tilde{\eta}_{D,p} = D'(p) \frac{p}{D(p)} = -f(p) \frac{p}{1 - F(p)}, \quad (\text{A36})$$

which implies in the normal case

$$\tilde{\eta}_{D,p}^{IM} - \tilde{\eta}_{D,p}^{MM} \propto - \underbrace{\left(\frac{\phi(\frac{p-\mu_\omega^{IM}}{\sigma_\omega})}{1 - \Phi(\frac{p-\mu_\omega^{IM}}{\sigma_\omega})} - \frac{\phi(\frac{p-\mu_\omega^{MM}}{\sigma_\omega})}{1 - \Phi(\frac{p-\mu_\omega^{MM}}{\sigma_\omega})} \right)}_{\text{level effect} \geq 0} \leq 0 \quad (\text{A37})$$

The inverse Mills ratio and the difference in the inverse Mills ratio are always positive by the log concavity of the normal distribution ensuring non-negativity. Thus consumers are more price elastic in immature markets. $\square$

A.4.2. Cost Expansion: General Results

Section 2.3 establishes cost expansion under normality. The following results generalise these to arbitrary marginal distributions under the conditions of Section 2.4.

LEMMA A3 (Adverse Selection In Levels). *When there is no re-sorting ($C = C^{IM} = C^{MM}$) marginal costs are higher at any price. However, when re-sorting occurs, effects on marginal cost are ambiguous.*

PROOF OF LEMMA A3. Analogously to the proof of Proposition 4

$$F_{\pi|\omega}^{IM}(\pi|\omega) \succeq F_{\pi|\omega}^{MM}(\pi|\omega) \longrightarrow MC^{IM}(p) \geq MC^{MM}(p) \forall p. \quad (\text{A38})$$

As the unconditional first order stochastic dominance of private valuations in mature and immature markets implies $F_\omega^{MM}(\omega) \leq F_\omega^{IM}(\omega)$, (A38) holds if and only if the copulas satisfy

$$C_2^{IM}(u, a) \leq C_2^{MM}(u, b) \forall a \geq b \text{ and } \forall u. \quad (\text{A39})$$

This is trivially satisfied under identical concave copulas ($C = C^{IM} = C^{MM}$).

With re-sorting, since $\int_0^1 C_2^{IM}(u, v) dv = \int_0^1 C_2^{MM}(u, v) dv = u$, the difference $C_2^{IM}(u, \cdot) - C_2^{MM}(u, \cdot)$ must change sign, so the ordering of $C_2^{IM}(u, a)$ and $C_2^{MM}(u, b)$ is not determined for arbitrary $a \geq b$, and the sign of $MC^{IM}(p) - MC^{MM}(p)$ is ambiguous. $\square$

To recover a marginal cost-slope definition of increased adverse selection, the notion of re-sorting must be strengthened. Specifically, it requires additionally satisfies a monotone regression dependence ordering.⁶ This condition is only needed to order

⁶Specifically, $\tilde{g}_u^{IM}(\tilde{g}_u^{MM}(v))$ is increasing in u for all $v \in (0, 1)$, where $g_x(y) = \partial C(x, y) / \partial x$ and $\tilde{g}_x \equiv g_x^{-1}$. Under normality this reduces to Assumption 4.

the *slope* of the marginal cost curve; the equilibrium results of Section 2.3 require only the weaker conditions on average costs established above.

LEMMA A4 (Increased Adverse Selection in Rank Space). *Consumers are adversely selected in both mature and immature markets. If re-sorting between markets satisfies the following additional assumption:*

- i. [Monotone Regression Dependence Ordering] $\tilde{g}_u^{IM}(g_u^{MM}(v))$ is increasing in u for all $v \in (0, 1)$, where $g_x(y) \equiv \frac{\partial C(x, y)}{\partial x}$ is the conditional distribution of V given $U = x$, and $\tilde{g}_x \equiv g_x^{-1}$ is its generalized inverse.

Then consumers are more adversely selected in immature markets in willingness-to-pay rank space. Price-space slope comparisons additionally depend on the marginal distribution of willingness-to-pay.

PROOF OF LEMMA A4. The proof in the general case follows from a direct application of Proposition 2.1 in Fang and Joe (1992). The result orders the regression of cost ranks on willingness-to-pay ranks under the stated monotone regression dependence ordering. Since this is a copula restriction, it delivers an ordering in rank space; transforming this ordering into derivatives with respect to price additionally involves the marginal density $f_\omega^m(p)$. $\square$

PROPOSITION A2 (Cost Expansion). *Average costs are increasing in price and early adopters are higher cost on average, $AC^{IM}(p) \geq AC^{MM}(p)$.*

PROOF OF PROPOSITION A2. The within-market ordering of average costs established in Proposition 4 Part (ii) relies on condition (ii) of that proposition – the corner-set monotonicity of the copula. As noted in the accompanying corollary, this is the relevant primitive for the cross-market comparison here. The necessary and sufficient condition for stochastic dominance of the truncated conditional distributions (see the proof of Proposition 4 and equation (A9)) is:

$$F_{\pi|\omega \geq p}(\pi|\omega \geq v_1) \leq F_{\pi|\omega \geq p}(\pi|\omega \geq v_2) \quad \forall v_1 \geq v_2 \text{ and } \forall \pi \quad (\text{A40})$$

$$\frac{u - C(u, F_\omega(v_1))}{1 - F_\omega(v_1)} \leq \frac{u - C(u, F_\omega(v_2))}{1 - F_\omega(v_2)} \quad \forall v_1 \geq v_2 \text{ and } \forall u. \quad (\text{A41})$$

The analogous condition to order average costs between markets is:

$$F_{\pi|\omega \geq p}^{IM}(\pi|\omega \geq v) \leq F_{\pi|\omega \geq p}^{MM}(\pi|\omega \geq v) \quad \forall v, \pi \quad (\text{A42})$$

$$\frac{u - C^{IM}(u, F_{\omega}^{IM}(v))}{1 - F_{\omega}^{IM}(v)} \leq \frac{u - C^{MM}(u, F_{\omega}^{MM}(v))}{1 - F_{\omega}^{MM}(v)} \quad \forall v, \forall u \quad (\text{A43})$$

As $F_{\omega}^{MM}(\omega) \leq F_{\omega}^{IM}(\omega)$, established in Proposition A1, the following is true for all copulas satisfying condition (ii) of Proposition 4 (see the proof above)

$$\frac{u - C^{MM}(u, F_{\omega}^{IM}(v))}{1 - F_{\omega}^{IM}(v)} \leq \frac{u - C^{MM}(u, F_{\omega}^{MM}(v))}{1 - F_{\omega}^{MM}(v)} \quad \forall v \text{ and } \forall u, \quad (\text{A44})$$

and by the concordance ordering in Assumption 3

$$\frac{u - C^{IM}(u, F_{\omega}^{IM}(v))}{1 - F_{\omega}^{IM}(v)} \leq \frac{u - C^{MM}(u, F_{\omega}^{IM}(v))}{1 - F_{\omega}^{IM}(v)} \quad \forall v \text{ and } u \in [0, 1], \quad (\text{A45})$$

which completes the proof. $\square$

The sufficient primitive for Proposition A2 is weaker than Assumption 3. The proof shows that $AC^{IM}(p) \geq AC^{MM}(p)$ follows directly from

$$F_{\pi|\omega \geq p}^{IM}(\pi|p) \leq F_{\pi|\omega \geq p}^{MM}(\pi|p) \quad \forall p, \pi, \quad (\text{A46})$$

i.e., $\pi|(\omega^{IM} \geq p) \succeq_{FOSD} \pi|(\omega^{MM} \geq p)$ for all p — upper-tail cost dominance across markets. Assumption 3 is sufficient for this but not necessary: it delivers the condition by combining demand contraction with the within-market corner-set property, but the cross-market ordering could in principle hold without global concordance of the copulas. Concordance is retained as the maintained assumption because it also delivers Proposition A3 directly, providing a unified condition for both price-space and quantity-space comparisons.

PROPOSITION A3 (Cost Expansion Fixing Demand). *Average Costs as a function of quantity are larger in immature markets, $AC^{IM}(q) \geq AC^{MM}(q)$. Marginal Costs rotate around a point and are higher at low quantities, but smaller at high quantities, thus $MC^{IM}(q) \geq MC^{MM}(q)$ for all $q \leq \tilde{q}$ and vice versa. Absent re-sorting, both cost curves are identical.*

Moreover, the adverse selection wedge is higher at any quantity demanded: $|AC^{IM}(q) - MC^{IM}(q)| \geq |AC^{MM}(q) - MC^{MM}(q)|$.

PROOF OF PROPOSITION A3. Note that the copula defines *rank* dependence. Consequently, they are invariant under strictly monotone transformations, including the inverse demand transformation considered here. Thus, to order average costs given quantity, q , we have the following condition:

$$\frac{u - C^{IM}(u, 1 - q)}{q} \leq \frac{u - C^{MM}(u, 1 - q)}{q} \quad \forall q \in (0, 1] \text{ and } u \in [0, 1], \quad (\text{A47})$$

which simplifies to

$$C^{IM}(u, 1 - q) \geq C^{MM}(u, 1 - q) \quad \forall q \in (0, 1] \text{ and } u \in [0, 1]. \quad (\text{A48})$$

This only depends on the change in sorting ($C^{IM} \neq C^{MM}$) and holds with equality under no re-sorting ($C = C^{IM} = C^{MM}$). The stated Proposition then follows from the concordance ordering in Assumption 3.

Equation (A39) gives an identical condition for marginal costs in terms of the conditional density which corresponds to stochastic dominance

$$C_2^{IM}(u, 1 - q) \leq C_2^{MM}(u, 1 - q) \quad \forall q \in (0, 1] \text{ and } u \in [0, 1]. \quad (\text{A49})$$

This cannot be satisfied as stochastic dominance for all conditional distributions implies stochastic dominance of the unconditional distribution. This violates assumption 1.

Note that at $q = 0$ marginal costs are equal to average costs. Then, the monotonicity of marginal costs and the results of Lemma A4 imply a single crossing property which completes the proof of rotation.

The rotation of the marginal cost curve and Lemma A4 imply the derivative of the marginal cost curve is also ordered in quantity space

$$\frac{\partial MC^{IM}(q)}{\partial q} \leq \frac{\partial MC^{MM}(q)}{\partial q} < 0, \quad (\text{A50})$$

and we can express the wedge as

$$W(q) = \frac{1}{q} \int_0^q (MC(t) - MC(q)) dt = - \int_0^q MC'(t) \frac{t}{q} dt. \quad (\text{A51})$$

Combining them gives $|AC^{IM}(q) - MC^{IM}(q)| \geq |AC^{MM}(q) - MC^{MM}(q)|$. □

A.4.3. Proofs for Imperfect Competition

PROOF OF LEMMA 1. I omit the subscript j for convenience. Let $S(q)$ denote the supply schedule in a market,

$$S(q) = MC(q) + \frac{1}{\eta_D(q)} \quad (\text{A52})$$

Excess supply is given by $ES(q) = p(q) - S(q)$ and $ES^{MM}(q^{MM}) = 0$ by definition. Then

$$ES^{IM}(q^{MM}) = p^{IM}(q^{MM}) - \left(MC^{IM}(q^{MM}) + 1/\eta_D^{IM}(q^{MM}) \right) \quad (\text{A53})$$

$$\leq p^{MM}(q^{MM}) - \left(MC^{IM}(q^{MM}) + 1/\eta_D^{IM}(q^{MM}) \right), \quad (\text{A54})$$

by the assumption of the demand contraction. Under the restriction on quantities and elasticities stated in the lemma we have

$$p^{MM}(q^{MM}) - \left(MC^{IM}(q^{MM}) + 1/\eta_D^{IM}(q^{MM}) \right) \quad (\text{A55})$$

$$= \underbrace{p^{MM}(q^{MM}) - \left(MC^{MM}(q^{MM}) + 1/\eta_D^{MM}(q^{MM}) \right)}_{=ES^{MM}(q^{MM})=0} \quad (\text{A56})$$

$$+ \underbrace{\left(MC^{MM}(q^{MM}) - MC^{IM}(q^{MM}) \right)}_{\leq 0} + \underbrace{\left(1/\eta_D^{MM}(q^{MM}) - 1/\eta_D^{IM}(q^{MM}) \right)}_{\leq -(MC^{MM}(q^{MM}) - MC^{IM}(q^{MM}))} \leq 0,$$

which establishes excess supply in immature markets at the mature market equilibrium. By the maintained single-crossing condition on the immature-market excess-supply schedule, the zero of $ES^{IM}(q)$ must occur weakly to the left of q^{MM} . Therefore $q_j^{MM} \geq q_j^{IM}$. It follows that $p_j^{MM} \leq p_j^{IM}$ by the rotation of the marginal cost curve and restriction on elasticities whenever the supply curve is monotone.

As the immature state both depresses inverse demand and raises average selection costs, the available profit margin at any relevant quantity is depressed, making the monopoly strictly worse off.

□

Appendix B. Additional Results for Section 3

B.1. Proofs for Section 3

Proof of the first part. In immature markets let $-C < 0$ denote the loss when both firms enter and $\Pi_M^{IM} < 0$ denote the monopolist profit, respectively. Negative profits are consistent with average cost pricing because the market unravels in immaturity. Equivalently Π_M^{MM} denotes the monopolist profit and $\Pi^{MM} = 0$ denotes the (competitive) profit in mature markets.

	enter	exit		enter	exit
enter	(0, 0)	($\Pi_M^{MM}, 0$)	enter	(-C, -C)	($\Pi_M^{IM}, 0$)
exit	(0, Π_M^{MM})	(0, 0)	exit	(0, Π_M^{IM})	(0, 0)
(a) Mature Market			(b) Immature Market		

TABLE A1. Second Period Payoff Matrix

Payoffs in the second subgame when at least one firm entered in the first period are given in Panel A of Table A1.

In a mature market that does not fully unravel $\Pi_M^{MM} > 0$ and entering is a profitable deviation for both firms if neither enter, thus this cannot be a nash equilibrium of the subgame. There is no profitable deviation from the symmetric strategy profile (enter, enter) as each firm also earns zero under unilateral deviation.

Next, consider the second subgame when neither firm entered in the first period (Panel B). When $\Pi_M^{IM} < 0$, exiting is a strictly dominant strategy and there is a unique equilibrium.

Taking these contingent strategy profiles as given, the normal form representation of the first subgame is in Table A2. As the market is immature and firms earn zero equilibrium profits in mature markets, these payoffs are identical to the immature market second period subgame and exit is strictly dominant.

	enter	exit
enter	($-C + 0, -C + 0$)	($\Pi_M^{IM} + 0, 0 + 0$)
exit	($0 + 0, \Pi_M^{IM} + 0$)	($0 + 0, 0 + 0$)

TABLE A2. First Period

Therefore, the following symmetric strategy profile is an SPE:

- First period: Play *exit*
- Second period: Play *exit* if no firm has entered, otherwise *enter*

and in equilibrium no firm enters in either period.

Proof of second part. When the market is immature in the second period, the SPE has firms playing strictly dominant strategies. Therefore, only refinements for the mature second period subgame need to be considered.

As this is a two player, two action game, the set of trembling hand perfect equilibria is equivalent to the set of equilibria in which no player plays a weakly dominated strategy. The equilibria considered above (enter, enter) when mature satisfies this. However, the asymmetric strategy profiles in the second round, either (entry, exit) or (exit, entry), do not because entering weakly dominates exiting.

This leaves a unique equilibrium surviving the trembling-hand refinement in this subgame. The same argument can be applied in the same fashion to the first period delivering a unique equilibrium of the extensive form game. $\square$

B.2. Generalizing the Stylized Model

Here I extend the two-firm two-period model from Section 3 to a richer dynamic setting. I show that the non-entry equilibrium extends to a dynamic model with partial cross-subsidization from future periods and with stochastic maturation. The difference is that, in the dynamic model, the proof relies on full continuation values. In addition, I extend it to a multi-firm framework and discuss the crucial features of the modelling environment that deliver uniqueness.

B.2.1. Multi-period Model: Stochastic Maturation with Partial Appropriability

The model is in discrete time and two firms discount the future at $\delta \in (0, 1)$. The market state is immature (*IM*) or mature (*MM*), it begins immature and maturity is an absorbing state. Each period firms simultaneously choose *enter* or *exit*, with per-period payoffs given by the reduced form of the entry-and-pricing stage of Section 3. Immature flow payoffs are as in the baseline: a sole entrant earns $\Pi_M^{IM} < 0$, two entrants each earn $-C < 0$, and a non-entrant earns 0. The mature stage game and the immature flow payoffs appear in Table A3.

If at least one firm enters and sells a positive quantity, the market becomes mature next period with probability $\lambda \in (0, 1]$ and remains immature with complementary

	enter	exit		enter	exit
enter	(0, 0)	$(\Pi_M^{MM}, 0)$	enter	$(-C, -C)$	$(\Pi_M^{IM}, 0)$
exit	$(0, \Pi_M^{MM})$	(0, 0)	exit	$(0, \Pi_M^{IM})$	(0, 0)
(a) Mature Market			(b) Immature Market (flow payoffs)		

TABLE A3. Stage payoffs

probability $1 - \lambda$. Absent entry it remains immature, so that maturation relies on entry as in Section 3. Deterministic maturation is the case $\lambda = 1$. As in the baseline, the mature market is competitive: its unique trembling-hand perfect outcome is $(enter, enter)$ with profit $\Pi^{MM} = 0$.⁷

A firm that is the *sole* entrant in the period the market transitions captures a fraction $\Lambda \in [0, 1]$ of the mature monopoly profit, $\Lambda \Pi_M^{MM}$. I model this as a one-time transition rent, after which the mature market is competitive, if both firms are active at the transition this rent is competed away.⁸

Using recursion, the value to a monopolist that enters every immature period until the market matures is given by

$$W = \Pi_M^{IM} + \delta [\lambda \Lambda \Pi_M^{MM} + (1 - \lambda) W] = \frac{\Pi_M^{IM} + \delta \lambda \Lambda \Pi_M^{MM}}{1 - \delta(1 - \lambda)}. \quad (A57)$$

Because the immature state is stationary, this single-agent problem is solved by entering in every period or in none, so the value of being a sole potential entrant is $\max\{0, W\}$. Define the cross-subsidization threshold

$$\Lambda^* = \frac{-\Pi_M^{IM}}{\delta \lambda \Pi_M^{MM}} = \frac{|\Pi_M^{IM}|}{\delta \lambda \Pi_M^{MM}}, \quad \text{so that} \quad W \leq 0 \iff \Lambda \leq \Lambda^*. \quad (A58)$$

The condition “a monopolist cannot earn positive profits in the immature market” of

⁷Resolving the mature market by its trembling-hand perfect outcome parallels the baseline, where the mature market is the terminal stage, and isolates the role of immaturity by ruling out collusive continuations sustained by the folk theorem in the absorbing mature phase. Equivalently, attention is restricted to Markov strategies once mature.

⁸One micro-foundation for $\Lambda > 0$ is a reaction lag. If the rival enters the matured market with per-period hazard $\rho \in (0, 1]$, a sole mover collects the mature monopoly flow until the rival reacts, so that $\Lambda = \frac{1-\delta}{1-\delta(1-\rho)}$. This is decreasing in the speed of reaction ρ and lies in $[1 - \delta, 1)$ for $\rho \in (0, 1]$. Thus the reaction-lag parametrization leaves a positive interim rent for fixed $\delta < 1$. The baseline case $\Lambda = 0$ corresponds instead to immediate Bertrand/free-entry competition upon maturity, or equivalently to ruling out any interim monopoly rent.

Section 3 thus generalizes from $\Pi_M^{IM} < 0$ to $W \leq 0$, with Λ subsumed into payoff W .

PROPOSITION A4. *Suppose $\Pi_M^{IM} < 0$, $-C < 0$, and $\Pi_M^{MM} > 0$. If a monopolist cannot earn positive profits in the immature market, i.e. $W \leq 0$ (equivalently $\Lambda \leq \Lambda^*$), there exists a subgame perfect equilibrium in which no firm enters in any period. If the inequality is strict, $W < 0$ (equivalently $\Lambda < \Lambda^*$), this is the unique extensive-form trembling-hand perfect equilibrium.*

Proof of the first part. Consider the symmetric profile

- in any immature period, play *exit*;
- in the mature market, play *enter*.

In the mature market this is the subgame of Panel A of Table A3: (*enter*, *enter*) is a Nash equilibrium and no unilateral deviation is profitable, so the profile is sequentially rational. In any immature period, suppose the rival exits in every immature period. Any strategy of firm i that ever enters receives single-agent payoff $\max\{0, W\} = 0$, while exiting forever attains 0; hence *exit* is a best response. The profile is therefore a subgame perfect equilibrium, and no firm enters along the path.

Proof of the second part. Take $\Lambda < \Lambda^*$, so that $W < 0$ and $\Pi_M^{IM} + \delta\lambda\Lambda\Pi_M^{MM} < 0$ by (A57).

Consider equilibria satisfying the trembling-hand refinement described above, which selects the competitive mature-market continuation. In any such equilibrium, each firm's continuation value at every immature node is zero. It cannot be negative, since exiting in every period guarantees 0.

For the converse, let $\bar{V} \geq 0$ denote the supremum of the continuation value across immature nodes and firms. At any immature node, *exiting* yields at most $\delta\bar{V}$. If the rival exits the market stays immature and the continuation is at most $\bar{V}$; if the rival enters, the market matures with probability λ and the exiting firm then receives the refinement-selected mature-market continuation payoff of 0, and with probability $1 - \lambda$ the continuation is at most $\bar{V}$. *Entering* yields at most

$$\max\{\Pi_M^{IM} + \delta\lambda\Lambda\Pi_M^{MM}, -C\} + \delta(1 - \lambda)\bar{V} < \delta\bar{V},$$

because the leading term is negative and $\delta(1 - \lambda)\bar{V} \leq \delta\bar{V}$. Hence every immature continuation value is at most $\delta\bar{V}$, so $\bar{V} \leq \delta\bar{V}$, which forces $\bar{V} = 0$. Thus every immature

continuation value is zero in any equilibrium satisfying the refinement. Hence the non-entry equilibrium is the unique refined subgame-perfect equilibrium.

With all continuation values equal to zero, the immature stage game is Table A4, which is the immature subgame of the two-period game with $\Pi_M^{IM} < 0$ replaced by $\Pi_M^{IM} + \delta\lambda\Lambda\Pi_M^{MM} < 0$. As both this quantity and $-C$ are negative, *exit* is strictly dominant, so no firm enters in any immature period in any subgame perfect equilibrium. Consequently, they do not enter in any trembling-hand perfect equilibrium.

	enter	exit
enter	$(-C, -C)$	$(\Pi_M^{IM} + \delta\lambda\Lambda\Pi_M^{MM}, 0)$
exit	$(0, \Pi_M^{IM} + \delta\lambda\Lambda\Pi_M^{MM})$	$(0, 0)$

TABLE A4. Immature stage game in any subgame perfect equilibrium

The market therefore never matures along the equilibrium path. As above the mature subgame is a two-player, two-action game in which *enter* weakly dominates *exit*, so the trembling-hand refinement uniquely selects $(enter, enter)$ and the asymmetric profiles do not survive.

Combining, the no-entry profile of the first part is the unique extensive-form trembling-hand perfect equilibrium. $\square$

Several features are worth noting. By (A58), Λ^* is decreasing in both λ and δ . Slower or more uncertain maturation enlarges the no-entry region, so stochastic maturation *strengthens* deterrence relative to the deterministic baseline, while greater patience shrinks it as cross-subsidization is more effective at funding earlier losses. The threshold is independent of any financing constraint, so an arbitrarily well-capitalized entrant still stays out whenever $\Lambda \leq \Lambda^*$. The binding constraint is long-run profitability, not capital, which is the precise sense in which deep pockets do not by themselves induce entry. Finally, since the strict dominance of *exit* holds period-by-period, both existence and uniqueness backward-induct through any finite horizon.

B.2.2. Multi-firm Model with Differentiated Products

With homogeneous-goods Bertrand competition the number of firms is immaterial: mature-market profits are competed to zero whenever two or more firms are active, so nothing in the equilibrium of the entry game depends on N . A natural extension is to allow within-period competition to lead to positive profits. Assumption A4 is the reduced form of imperfect substitutability among firms' contracts, firms retain positive margins

that decline as additional firms crowd the market. This naturally arises under spatial competition à la Hotelling or Salop, or differentiated-products (e.g. logit) demand⁹ I also maintain that the immature market is unprofitable at any structure, $\pi^{IM}(n) < 0$ for all $n \leq N$, consistent with the adverse selection primitive in Section 3.

ASSUMPTION A4 (Positive Mature Profits). $\pi^{MM}(n)$ is strictly positive and non-increasing in n on $\{1, \dots, N\}$, with $\pi^{MM}(1) = \Pi_M^{MM}$.

The substantive finding is that doing so makes the no-entry result *more* likely as N grows—a larger pool of potential entrants shrinks their ability to earn sufficient reward in the mature market, so the market is increasingly unlikely to mature.

As above N firms discount at $\delta \in (0, 1)$ and in each period simultaneously choose *enter* or *exit*. When n firms are active, each earns the per-firm flow $\pi^{IM}(n)$ if the market is immature and $\pi^{MM}(n)$ if it is mature; these generalize the baseline, with $\pi^{IM}(1) = \Pi_M^{IM}$, $\pi^{IM}(2) = -C$, and $\pi^{MM}(1) = \Pi_M^{MM}$. As before, if at least one firm enters and sells a positive quantity the market matures next period with probability $\lambda \in (0, 1]$, independently of the number of entrants, and absent entry it stays immature. A firm that is the *sole* entrant when the market transitions collects the one-time rent $\wedge \Pi_M^{MM}$ in that period.

Under Assumption A4 entering the mature market yields $\pi^{MM}(n) > 0$ against any number of rivals while exiting yields 0, so *enter* is strictly dominant for each firm. All N firms enter and each attains the continuation value

$$V^{MM} = \frac{\pi^{MM}(N)}{1 - \delta} > 0. \quad (\text{A59})$$

This differs from the two-firm variant and no longer requires a trembling-hand refinement.

Consider the value of a lone-entrant during every immature period until maturation. This gives

$$W_N = \frac{\Pi_M^{IM} + \delta\lambda(\wedge \Pi_M^{MM} + V^{MM})}{1 - \delta(1 - \lambda)} = W + \frac{\delta\lambda}{1 - \delta(1 - \lambda)} \cdot \frac{\pi^{MM}(N)}{1 - \delta}, \quad (\text{A60})$$

where W is the value in equation (A57). The value equals the Bertrand value W plus the discounted permanent rent V^{MM} . The denominator is positive, so W_N inherits the

⁹The homogeneous-goods Bertrand benchmark above is the excluded limit $\pi^{MM}(n) = \Pi^{MM} = 0$ for $n \geq 2$. Assumption A4 is exactly the departure that makes mature profits, and hence the entry incentive, sensitive to N .

sign of the per-period gamble $\Pi_M^{IM} + \delta\lambda(\Lambda\Pi_M^{MM} + V^{MM})$. The optimal entry decision of a lone-entrant is bang-bang and the value of being a lone potential entrant is $\max\{0, W_N\}$.

PROPOSITION A5. *Under Assumption A4 and $\pi^{IM}(n) < 0$ for all $n \leq N$, then, if $W_N \leq 0$, there is a Markov perfect equilibrium in which no firm enters while the market is immature and all enter once it is mature. If $W_N < 0$, this is the unique Markov perfect equilibrium.*

Proof. The mature phase is immediate from strictly dominated strategies. Take the strategy “exit while immature, enter while mature.” Given rivals exit, a unilateral entrant faces the stationary single-agent problem with value $\max\{0, W_N\} = 0$, while exiting forever attains 0. By assumption, no deviation is profitable. This establishes existence.

For *uniqueness*, suppose $W_N < 0$, equivalently $g \equiv \Pi_M^{IM} + \delta\lambda(\Lambda\Pi_M^{MM} + V^{MM}) < 0$. Restrict to Markov strategies and let $V_i^{IM} \geq 0$ be firm i ’s continuation value at the immature state. This is nonnegative because perpetual exit guarantees 0. Fix i and condition on the number $k \in \{0, \dots, N-1\}$ of rivals that enter.

- For $k = 0$ entering yields $\Pi_M^{IM} + \delta[\lambda(\Lambda\Pi_M^{MM} + V^{MM}) + (1-\lambda)V_i^{IM}]$, while exiting leaves the market immature and yields δV_i^{IM} . The difference in payoff from the deviation is $g - \delta\lambda V_i^{IM} \leq g < 0$.
- For $k \geq 1$, the market matures with probability λ irrespective of i ’s entry decision. Potential entrant i also earns no transition rent since a rival is always active. The continuation value from entering and not entering coincide and the difference in payoffs from deviating is $\pi^{IM}(k+1) < 0$.

Hence *exit* is strictly dominant at the immature state, so every Markov perfect equilibrium has all firms exiting while immature. Combined with the mature phase, the no-entry profile is the unique Markov perfect equilibrium by strict dominance. $\square$

Note that the permanent rent enters (A60) only through $\pi^{MM}(N)$, which is non-increasing in N , so W_N is non-increasing in the number of firms. A larger pool of potential entrants weakens mature market profitability, thereby weakening incentive to enter in immaturity. Consequently, there is thus a cutoff in N below which $W_N > 0$ and the market is provided and above which $W_N < 0$ and no firm enters. More competition therefore makes the market less likely to form, linking to the policy implication of the simple model.

Proposition A5 uniqueness in Markov strategies, not all SPE. As the mature market is profitable it is possible to sustain early entry, and cooperative provision of maturation,

via trigger strategies which the Markov assumption rules out. Notably, since V^{MM} falls in N , even this is unsustainable for large N .

Appendix C. Additional Discussion of Section 4

Testimonials. To highlight the widespread use of the Annuity Shopper quotes I reproduced the following testimonial from the July 2005 issue. The testimonial is addressed to Hersh Stein (the publisher):

Dear Hersh:

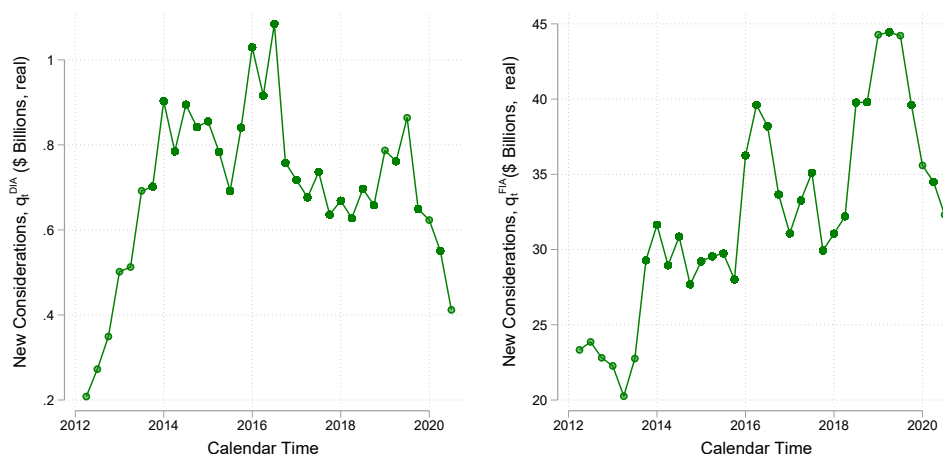
Thank you for providing the WebAnnuties packet and the Annuity Shopper. It is the best source of information I have found on the subject of annuities. And a special thanks for calling and answering my questions.

As an attorney who frequently represents clients regarding investments, retirement and estate planning I will highly recommend you for their annuity needs. As I mentioned I have several annuities now and will continue to add more to my own retirement portfolio.

I have recommended you and your company to other estate planners and financial advisors.

Very truly yours, David D. Dunakey

Similar testimonies are frequently published.



(A) Deferred Income Annuities

(B) Other Fixed Income Annuities

FIGURE A2. Flow Consideration In The Deferred Income Annuities Market

Sales of Annuity Products. Figure A2 shows how the size of this market has evolved over time with the left panel displaying new deferred income annuity sales. At the start

of this period the market for deferred income annuities was small, totaling around \$200 million and accounting for less than 1% of the overall fixed income annuity market. Subsequently, however, the market expanded rapidly between 2012 and 2014, with the real value of new considerations more than quadrupling in size. The right panel shows sales of all other fixed income annuity products, the most similar alternative product in the annuity space. Comparing the two panels suggests an element of common cyclicalities, but growth in fixed income annuities is much smaller over the period.

Entrant Composition and Product Mix. The main text reports the number of distinct entrants and the overall number of firms operating in this market.

To better understand the composition of insurers in the market, I classify insurers using two indicators. First, I define a group of Mutual insurers consisting of Guardian Life, MassMutual, Mutual of Omaha, and New York Life. These insurers are grouped by their ownership structure. Second, I define a group of traditional-large insurers. This consists of American General, Lincoln National, MetLife, and Pacific Life, in addition to the mutual insurers. These are insurers with an established national presence and broad market scale. In subsequent robustness analysis I use both as proxies for the cross-subsidization incentives of large-multi-product insurers.

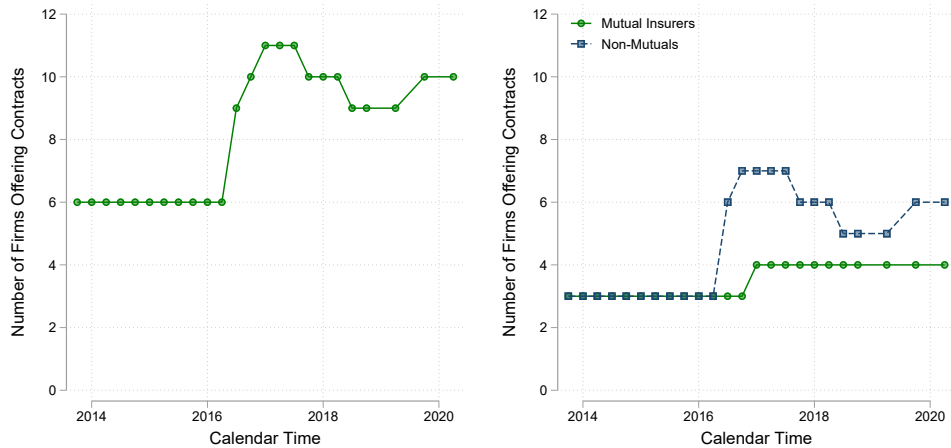
Figure A3 shows that the market is initially evenly split between mutual insurers and non-mutual insurers (Panel B), but that it is primarily composed of large-traditional insurers with a national presence (Panel C). The expansion of the market after 2016 is disproportionately non-mutual firms. Relative to the existing stock of entrants, the post 2016 operators are also more likely to be smaller insurers.

Table A5 shows that insurer offerings are highly structured. The majority of firms operating offer either a full product grid or a reduced core grid, rather than idiosyncratic subsets of contracts. The difference between these is that the core grid excludes long deferral annuities purchased at age 45 and deferred until 80 or 85. Most observed product-line changes are around this core set, especially the permanent dropping of long-deferral contracts or limited experimentation with short-deferral lines. Exit is rare and mostly reflects firms leaving the observed market rather than repeatedly cycling in and out. Overall, the table suggests that product availability is fairly stable once firms enter, with adjustments occurring through selective pruning of particular product lines. Although it is consistent with it, the exit from specific lines should not be taken as *prima facie* evidence of unravelling.

Insurer	2013	2014	2015	2016	2017	2018	2019	2020Q1
American Gen	Enter Full	✓	✓	✓	Drop $a = 45, d = \{80, 85\}$	✓	✓	✓
Brighthouse	×	×	×	×	Enter Core	✓	Exit Core	×
Foresters	×	×	×	Enter Full	✓	Exit Full	×	×
Guardian Life	Enter Full	✓	✓	✓	✓	✓	✓	✓
Integrity	×	×	×	×	×	×	Enter Full	✓
Lincoln Nat	×	×	×	Enter Full	✓	✓	✓	✓
Mass Mutual	Enter Core	✓	✓	Drop $a = 65, d = 70$	Add $a = 65, d = 70$	Drop $a = 65, d = 70$	✓	✓
Metlife	Enter Full	✓	✓	Drop $a = 45, d = \{80, 85\}$	✓	Exit Core	×	×
Mutual of Omaha	×	×	×	Enter Full	✓	✓	✓	✓
New York Life	Enter Full	✓	✓	✓	✓	✓	✓	✓
Pacific Life	×	×	×	Enter Core	✓	✓	✓	✓
Principal	×	×	×	Enter Core	✓	✓	✓	✓
Symetra	Enter Full	✓	✓	✓	✓	✓	✓	✓

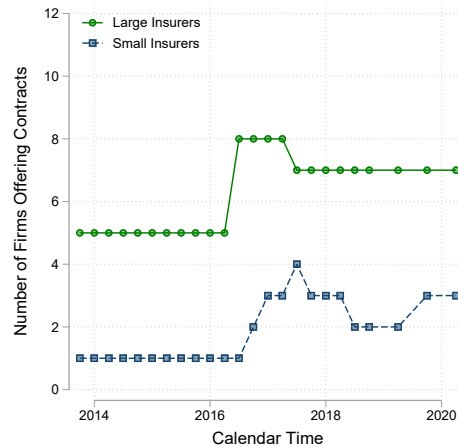
Notes: ✓ denotes an active insurer-year and × denotes inactivity. “Enter” and “Exit” denote the first observed entry or exit in the displayed sample. Product lines are defined by sex, purchase age, and annuity start age. “Full” denotes the full product grid. “Core” excludes the long-deferral purchase-age-45 lines with start ages 80 and 85. “ $a = 45, d = \{80, 85\}$ ” denotes those excluded long-deferral lines. “ $a = 65, d = 70$ ” denotes the purchase-age-65/start-age-70 product lines. The 2020 column includes Q1 only.

TABLE A5. Insurer Entry, Exit, and Product-Line Changes



(A) All Firms

(B) By Firm Legal Status



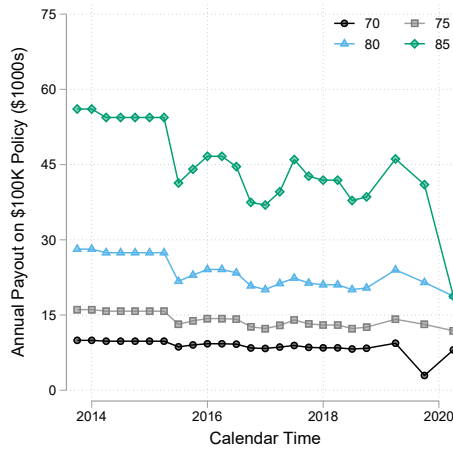
(C) By Firm Size

Notes: Panel A plots the number of distinct insurers quoting deferred income annuities each quarter. Panels B and C subdivide into mutual insurers vs non-mutual insurers (B) and large-traditional insurers vs smaller insurers (C) as described in the text. From *Annuity Shopper*, 2013Q3–2020Q1.

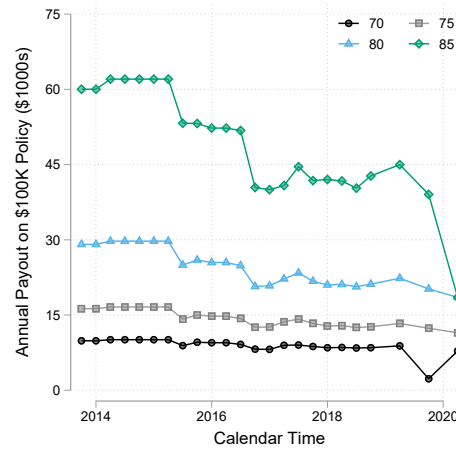
FIGURE A3. Composition of Firms in the Market for Deferred Income Annuities

Between and Within Insurer Payout Variation. Figure A5 repeats the exercise in Figure 5 for contracts offered by American General in addition to New York Life, highlighting similar movements across insurers.

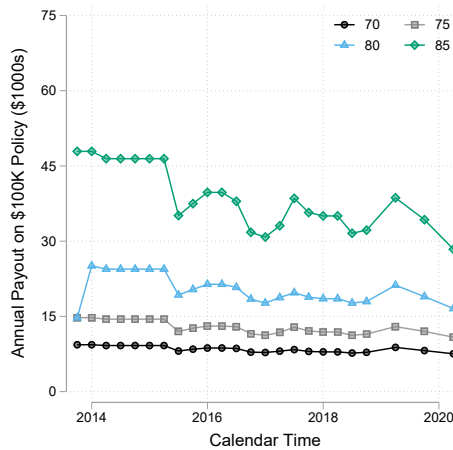
Figure A5 repeats the exercise in Figure 5 for contracts offered by New York Life, but conditioning on the deferral age and varying the purchase age. The left panel shows contracts for men and the right panel for women with each line corresponding to a different purchase age between 45 and 65. Again, contracts that offer payouts further



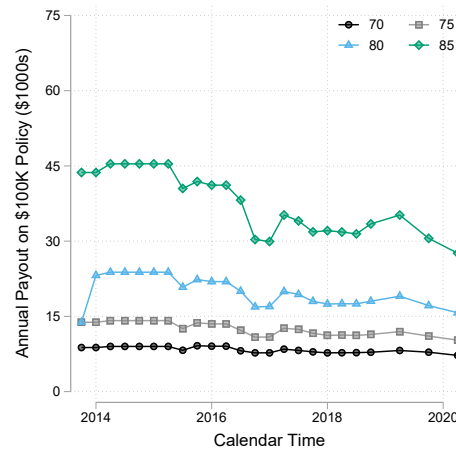
(A) American General (Men)



(B) New York Life (Men)



(C) American General (Women)



(D) New York Life (Women)

Notes: Annual payout per \$100,000 premium for single-life, non-refundable deferred income annuities purchased at age 65, by gender, and deferral age (70–85). From *Annuity Shopper*, 2013Q3–2020Q1.

FIGURE A4. Deferred Income Annuity Payouts at Purchase Age 65 By Insurer

into the future have higher payout amounts. Similarly, payouts for men who have lower expected longevity at all ages are larger than for women. Despite systematic differences by product characteristics, a notable feature of both Figures A4 and A5 is evidence of a market level trend in payouts over time for all contract terms and for both insurers. This reflects that the largest cost to the insurer is the opportunity cost of foregone investment (see Mitchell et al. 1999) as well as a changing pool of annuitants.

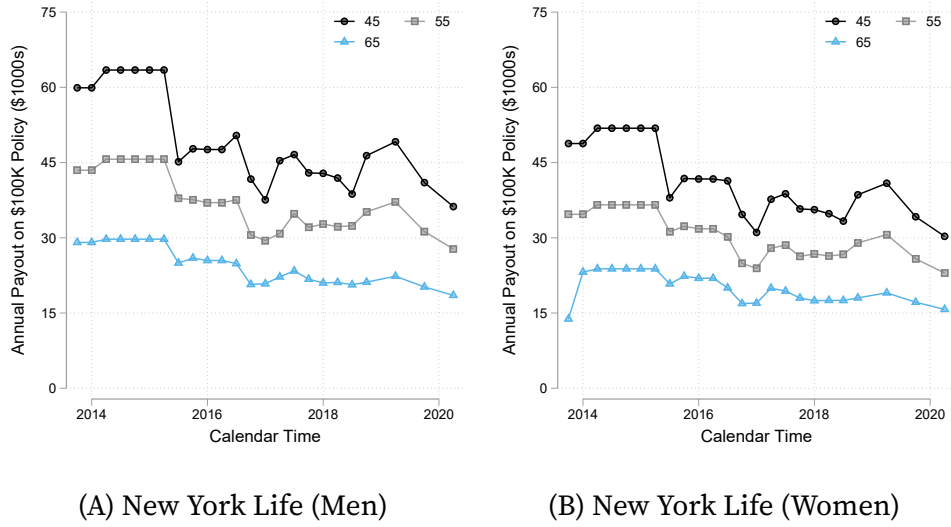


FIGURE A5. Average Payouts from Deferred Income Annuity Policies Beginning Payment at Age 80 By Gender and Purchase Age

C.1. Additional Empirical Results for Section 4

	Alternative Controls			Alternative Selection Measures			
	(1) Base	(2) Parametric Time Trend	(3) Policy FEs	(4) Levels	(5) IAM Life Table	(6) 10-Year Treasury	(7) Treasury Yield Curve
$\mathbb{1}[a > 45] \ln Q_t$	-0.380 (0.106)	-0.351 (0.093)	-0.376 (0.107)	-0.073 (0.022)	-0.102 (0.034)	-0.083 (0.035)	-0.015 (0.027)
Wild bootstrap p-value	0.004	0.005	0.004	0.021	0.027	0.076	0.609
Observations	5,058	5,058	5,058	5,104	5,104	5,104	5,104
R ²	0.703	0.550	0.711	0.806	0.675	0.787	0.744

Notes: Standard errors are two-way clustered at the Insurer and Time Period level with p-values constructed using the wild bootstrap (MacKinnon et al. 2021). Estimates reported for coefficient δ in equation (21). Implied selection is recovered using equation (20) and panel data collected from policy quotes printed in *Annuity Shopper* magazine. Aggregate quantities correspond to the cumulative real value of new deferred income annuity considerations collected from LIMRA and new considerations for all other fixed income annuity (FIA) products reported in Figure 4. All Specifications additionally control for purchase age indicators, insurer specific indicators, gender, deferral age indicators and time fixed effects (excluding (2) due to colinearity with the time specific aggregate demand for all other FIA products and cumulative sales).

TABLE A6. Alternative Specifications

Alternative Controls. Table A6 presents alternative specifications corresponding to the results in Table 1. The first column repeats the baseline estimate from the final specification with the full set of controls in Table 1. The second column replaces the non-parametric time trend with a controls for contemporaneous demand in all other fixed

income annuity markets and cumulative demand in the deferred income annuity market, the un-interacted $\ln Q_t$. The parameter estimate is almost unchanged, highlighting that identification does not rely on the specification of the control for aggregate market conditions.

Finally, the third column replaces the controls for policy characteristics with a finer set of interact-characteristic fixed effects. Each annuity policy is identified by an insurer-gender-deferral-purchase age specific identifier and this specification controls for fixed effects along this interacted dimension. This allows for more granular differences in how insurers market to or administer plans by gender and deferral age which may drive estimates of the effect of maturity due to differential entry and exit of insurers and policy combinations. The point estimate is almost identical to the baseline specification.

Alternative Selection and Mortality Tables. The fourth column replaces the dependent variable with selection measured as deviations in levels rather than logs. The magnitude of the estimate changes due to the alternative specification of the dependent variable, but the qualitative patterns remain. The fifth column uses life tables from the individual Annuitant Mortality (IAM) 2012 table, which is described in American Academy of Actuaries / Society of Actuaries (2011). As discussed by [Poterba and Solomon \(2021\)](#) this forecasts significant mortality improvement at odds with observed mortality improvements. Consequently, the overall money's worth measure of annuities is more positive and the log specification discards a large share of the observations where the implied NPV exceeds the cost to consumers. Nevertheless, results across columns 4 and 5, both using levels, suggest that there are similar qualitative findings whether deviations are measured from the pool of all potential consumers or the pool of typical annuitants.

Alternative Opportunity Cost of Capital. Finally, columns six and seven consider alternative assumptions for the insurer opportunity cost of capital. Contracts with different deferral-age and purchase-age combinations have different payout horizons, so if the assumed discount rate differs markedly from the insurer's actual cost of capital this may generate spurious variation across contract types. As with alternative life tables, I present results for the level of the outcome variable. In column six, the contract-date specific interest rate uses the 10-year zero-coupon Treasury yield from the [Gürkaynak et al. \(2007\)](#) fitted yield curve in place of the BBB corporate bond yield. This follows recent Department of Labor recommendations for defined contribution plan sponsors

(see [Poterba and Solomon 2021](#)). Despite the substantially different level and dynamics of this rate due to the underlying risk premium, the results remain statistically significant and qualitatively consistent with the baseline, confirming that the findings are not driven by the specific choice of corporate yield or the embedded credit spread. Moreover, they are very close to results in levels for the baseline selection measure which uses an analogous specification.

Horizon-Specific Discounting. Column seven discounts each cash-flow horizon at the corresponding zero-coupon rate from the full [Gürkaynak et al. \(2007\)](#) yield curve rather than applying a single rate to all horizons. This specification produces estimates close to zero. The natural concern is that horizon-specific discounting is the *correct* treatment of contracts with heterogeneous payout dates, in which case the attenuation would indicate that the baseline differential is itself an artifact of applying one rate across horizons over a period when the term structure moved. Three considerations point against that reading.

First, the horizon-specific risk-free curve is not simply a less variable version of the baseline rate—over the sample it moves differently from, and at times opposite to, the BBB corporate yield that proxies the insurer’s opportunity cost. Reflecting a time-varying credit spread, the relevant zero-coupon rates are imperfectly correlated with BBB. Discounting each horizon along this curve therefore does not preserve the variation in money’s worth that the baseline rate generates; it reweights cash flows by a rate that may diverge from insurers’ actual cost of capital, even at a constant maturity. This is consistent with the results in column (6).

Second, the differential does not depend on cross-horizon variation in the way the spurious discounting hypothesis requires. Holding the deferral age fixed and varying only purchase age (Table 3 in the main text) shuts down the horizon composition channel through which heterogeneous horizons and a common rate could otherwise manufacture a spurious age gradient. Attenuation at the longest horizon is consistent with compression: distant payouts are heavily discounted under any rate, so the longest-dated contracts carry the least recoverable variation. An artifact of applying a flat rate across heterogeneous horizons would, if anything, be *strongest* where horizons differ most.

Third, and most directly, the premise of the design does not rest on money’s worth at all. The subjective-belief results (Table 4, main text) establish that mortality-relevant private information rises with age. This evidence is entirely agnostic on discounting

choice. The column-seven result is therefore best read as a limitation of the money's-worth framework—when the discount-rate structure removes the cross-contract variation, the test loses power—rather than as evidence against selection. The baseline BBB specification reflects an opportunity cost relevant to insurers' actual pricing decisions and preserves that variation.

Disaggregated Results. Table A7 repeats this exercise for the results reported in Table 2 which allow for heterogeneous effects by purchase age. Once again the results are robust to alternative controls and definitions of the dependent variable. Point estimates for specifications with alternative controls are almost identical and alternative specifications of the dependent variable yield similar qualitative findings that echo results reported in Table 2. This shows that results are not driven by particular specification choices or by assumptions about parameters used to calculate the net present value of insurer payouts. With the exception of column seven, all specifications in both Tables A6 and A7 the estimated effects remain economically significant and statistically significant at conventional levels.

	Alternative Controls			Alternative Selection Measures			
	(1) Base	(2) Parametric Time Trend	(3) Policy FEs	(4) Levels	(5) IAM Life Table	(6) 10-Year Treasury	(7) Treasury Yield Curve
$\mathbb{1}[a = 55] \ln Q_t$	−0.258 (0.089)	−0.229 (0.082)	−0.255 (0.090)	−0.049 (0.018)	−0.067 (0.027)	−0.058 (0.027)	−0.009 (0.020)
$\mathbb{1}[a = 65] \ln Q_t$	−0.535 (0.140)	−0.505 (0.123)	−0.533 (0.142)	−0.102 (0.030)	−0.146 (0.045)	−0.115 (0.045)	−0.022 (0.036)
Wild bootstrap p-value, age-55	0.037	0.063	0.037	0.051	0.072	0.096	0.684
Wild bootstrap p-value, age-65	0.002	0.003	0.003	0.021	0.027	0.047	0.565
Wild bootstrap p-value, 55=65	0.000	0.000	0.001	0.042	0.051	0.062	0.537
Observations	5,058	5,058	5,058	5,104	5,104	5,104	5,104
R ²	0.716	0.563	0.724	0.812	0.684	0.791	0.744

Notes: Standard errors are two-way clustered at the Insurer and Time Period level with p-values constructed using the wild bootstrap (MacKinnon et al. 2021). Estimates reported for coefficient δ_{55} and δ_{65} in equation (23). Implied selection is recovered using equation (20) and panel data collected from policy quotes printed in *Annuity Shopper* magazine. Aggregate quantities correspond to the cumulative real value of new deferred income annuity considerations collected from LIMRA and new considerations for all other fixed income annuity (FIA) products reported in Figure 4. All Specifications additionally control for purchase age indicators, insurer specific indicators, gender, deferral age indicators and time fixed effects (excluding (2) due to colinearity with the time specific aggregate demand for all other FIA products and cumulative sales).

TABLE A7. Alternative Specifications by Private Information

C.1.1. Heterogeneity in Effects by Insurer

The preferred specification in the main analysis controls for insurer-specific fixed effects. This accounts for permanent differences in the incentives to cross-subsidize products or use them as loss-leaders. A concern is that heterogeneity in the incentive to use these products strategically by insurers may evolve over time in a way that is heterogeneous by purchase age, and therefore not captured by either time effects or insurer-specific fixed effects.

One version of this concern is that because the DIA market is dominated by mutual life companies and large national insurers there may be other form of value in getting existing customers to purchase “add-on” products or strengthen customer retention. This would imply the actuarially fair pricing benchmark is the wrong benchmark from the firm’s perspective. I investigate this using two proxies for multi-product insurers with this strategic incentive: those who are mutual insurers, and those who are traditional-large insurers operating in the national market (defined above).¹⁰

The baseline analysis allows for fixed deviations from actuarially fair pricing across insurers. I compare this specification with an enriched specification which allows the estimated pricing differentials to vary with insurer type. I report the results for the mutual insurers compared to non-mutual in Table A8 and the result for traditional-large insurers compared to the small insurers in Table A9.

In both cases, I first compare the estimates for the pooled specification, which estimates the differential relative to age 45 purchasers, with and without the additional interaction with insurer categorization. The estimated effect in the second column which allows for heterogeneous impacts across insurers is statistically indistinguishable from the baseline effect. Moreover, the null-effect of mutuals or large insurers is tightly estimated: ruling out even relatively small differences. The third and fourth columns of both tables then repeat this exercise for the specification that disaggregates age 55 and 65 effects. Again, the estimated differences across insurer type are economically and statistically insignificant. Taken together, the decline appears where the cross-product channel is least operative, indistinguishably from where it’s most operative. This is suggestive evidence against a cross-product or loss-leading channel that is *differential* across purchase-age over time.

¹⁰Note that non-mutual insurers are evenly split in both the pre- and post-2016 regimes while small insurers tended to enter later.

	Pooled		By Private Information	
	(1) Base	(2) Mutual Interacted	(3) Base	(4) Mutual Interacted
$\mathbb{1}[a > 45] \ln Q_t$	−0.380 (0.106)	−0.377 (0.104)		
$\mathbb{1}[\text{Mutual}] \times \mathbb{1}[a > 45] \ln Q_t$		−0.008 (0.021)		
$\mathbb{1}[a = 55] \ln Q_t$			−0.258 (0.089)	−0.254 (0.084)
$\mathbb{1}[\text{Mutual}] \times \mathbb{1}[a = 55] \ln Q_t$				−0.011 (0.022)
$\mathbb{1}[a = 65] \ln Q_t$			−0.535 (0.140)	−0.533 (0.140)
$\mathbb{1}[\text{Mutual}] \times \mathbb{1}[a = 65] \ln Q_t$				−0.005 (0.020)
Wild bootstrap p-value, Mutual $\times$ pooled	.	0.712	.	.
Wild bootstrap p-value, Mutual $\times$ age-55	.	.	.	0.644
Wild bootstrap p-value, Mutual $\times$ age-65	.	.	.	0.792
Observations	5,058	5,058	5,058	5,058
R ²	0.703	0.704	0.716	0.716

Notes: Standard errors are two-way clustered at the Insurer and Time Period level with p-values constructed using the wild bootstrap (MacKinnon et al. 2021). Columns (1) and (2) report estimates for coefficient δ in equation (21), enriched by an interaction with an indicator for mutual funds in column (2). Columns (3) and (4) report estimates for coefficient δ_{55} and δ_{65} in equation (23), enriched by an interaction with an indicator for mutual funds in column (3). Implied selection is recovered using equation (20) and panel data collected from policy quotes printed in *Annuity Shopper* magazine. All Specifications additionally control for purchase age indicators, insurer specific indicators, gender, deferral age indicators and time fixed effects.

TABLE A8. Heterogeneity in Decline Across Mutual Funds

C.2. Heterogeneity by Deferral Age

The main text shows results conditioning on specific payout horizons. However, it is not possible to replicate this analysis for all horizons due to the structure of the data. A potential threat to interpreting these results as evidence in favor of private-information is that the horizon differentials are driven by deferral age composition. Because deferral ages are fixed across purchase ages, older purchasers mechanically hold contracts closer to the onset of payouts. If the maturity–selection relationship varies systematically on this way, then the pooled differential in equation (21) could partly capture horizon rather than private information.

	Pooled		By Private Information	
	(1) Base	(2) Large-Traditional Interacted	(3) Base	(4) Large-Traditional Interacted
$\mathbb{1}[a > 45] \ln Q_t$	-0.380 (0.106)	-0.393 (0.105)		
$\mathbb{1}[\text{Trad}] \times \mathbb{1}[a > 45] \ln Q_t$		0.023 (0.024)		
$\mathbb{1}[a = 55] \ln Q_t$			-0.258 (0.089)	-0.275 (0.087)
$\mathbb{1}[\text{Trad}] \times \mathbb{1}[a = 55] \ln Q_t$				0.031 (0.028)
$\mathbb{1}[a = 65] \ln Q_t$			-0.535 (0.140)	-0.541 (0.139)
$\mathbb{1}[\text{Trad}] \times \mathbb{1}[a = 65] \ln Q_t$				0.012 (0.021)
Wild bootstrap p-value, Large-Traditional $\times$ pooled	.	0.710	.	.
Wild bootstrap p-value, Large-Traditional $\times$ age-55	.	.	.	0.714
Wild bootstrap p-value, Large-Traditional $\times$ age-65	.	.	.	0.642
Observations	5,058	5,058	5,058	5,058
R ²	0.703	0.704	0.716	0.717

Notes: Standard errors are two-way clustered at the Insurer and Time Period level with p-values constructed using the wild bootstrap (MacKinnon et al. 2021). Columns (1) and (2) report estimates for coefficient δ in equation (21), enriched by an interaction with an indicator for mutual funds in column (2). Columns (3) and (4) report estimates for coefficient δ_{55} and δ_{65} in equation (23), enriched by an interaction with an indicator for insurer size in column (3). Implied selection is recovered using equation (20) and panel data collected from policy quotes printed in *Annuity Shopper* magazine. All Specifications additionally control for purchase age indicators, insurer specific indicators, gender, deferral age indicators and time fixed effects.

TABLE A9. Heterogeneity in Decline Across Traditional Large Insurers

I therefore re-estimate equation (21) separately within each deferral age, isolating the differential effect of maturity across purchase ages within a single homogeneous contract. Table A10 reports the results: the first column reproduces the preferred pooled estimate of -0.380 , and the remaining columns estimate the differential within each deferral age from 65 to 85.

The differential is negative for every deferral age and significant until age 80, ranging from -0.506 to -0.272 . It is largest for the nearer-dated contracts and attenuates at the longest horizon, falling to -0.161 at deferral age 85. This is consistent with the documented payout horizon effects and compression of distant payouts.

	Deferral Age Specific Estimates					
	(1) Base	(2) d=65	(3) d=70	(4) d=75	(5) d=80	(6) d=85
$\mathbb{1}[a > 45] \ln Q_t$	−0.380 (0.106)	−0.467 (0.105)	−0.506 (0.146)	−0.384 (0.106)	−0.272 (0.113)	−0.161 (0.103)
Wild bootstrap p-value	0.004	0.002	0.002	0.003	0.015	0.262
Observations	5,058	761	1,123	1,146	1,021	1,007
R ²	0.703	0.775	0.607	0.662	0.655	0.669

Notes: Standard errors are two-way clustered at the Insurer and Time Period level with p-values constructed using the wild bootstrap (MacKinnon et al. 2021). Estimates reported for coefficient δ in equation (21). Implied selection described in equation (20) and panel data collected from policy quotes printed in *Annuity Shopper* magazine. All specifications additionally control for purchase age indicators.

TABLE A10. Fixed-Deferral Age Tests for Adverse Selection

Appendix D. Health and Retirement Survey: Data & Additional Results

I use the Health and Retirement Study (HRS), a representative panel of US households aged 50 and older surveyed biennially since 1992. The analysis draws on processed data from Foltyn and Olsson (2024), who harmonize subjective survival beliefs with demographic characteristics and mortality outcomes across waves 1992–2018.¹¹

The HRS elicits survival expectations with the survey instrument: “What is the percent chance that you will live to be [X] or more?” where the target age X is 10 years in the future and varies with respondent age (except 75 for respondents aged 65 or younger).

I rescale responses to the unit interval. The sample is restricted to ages 50–70 and to observations where 10-year mortality status is resolved—either through observed death or sufficient follow-up. I focus on this age range as individuals younger than 50 are all spouses of HRS main respondents, therefore not representative, and the 50-70 samples closely aligns to the age groups I consider for annuity purchase age. I focus on a subsample with resolved mortality: either dying or being observed at least 10 years later. This yields 109,316 person-wave observations.

Following Hurd and McGarry (2002), I compare realized mortality across individuals

¹¹The replication package is available directly from the HRS at <https://hrsdata.isr.umich.edu/data-products/hrs-replication-foltyn-olsson>.

with different subjective survival beliefs at different ages. I construct an indicator for 10-year mortality, $M_{i,t+10}$, and estimate the linear probability model

$$M_{i,t+10} = \beta_0 + \beta_{surv} \text{Belief}_{i,t} + \beta_{Age} \text{Age}_{i,t} + \delta \text{Belief}_{i,t} \times \text{Age}_{i,t} + \beta X_{i,t} + u_{i,t}, \quad (\text{A61})$$

with age centered at 50, so that β_{surv} captures the effect of survival beliefs at age 50 and δ measures how predictive accuracy varies by age.

D.1. Focal Point Answers

A potential concern is that older respondents provide less noisy answers due to mortality salience rather than possessing more private information. Average subjective mortality exhibits a “flatness” bias due to focal-point heaping at 0%, 50%, and 100% (see, e.g., Hamermesh 1985; Elder 2013; Foltyn and Olsson 2024).

Figure A6 plots focal response rates by age. The share giving focal answers is roughly constant between ages 50 and 70, fluctuating between 42% and 47% with only a small upwards trend. This rules out differential response quality as an explanation for the increasing predictive power of beliefs.

Table A11 provides a direct test. Column 1 reproduces the baseline specification from Table 4. Column 2 excludes focal responses entirely. The interaction coefficient strengthens from -0.005 to -0.007 , indicating that answers without heaping carry more information. The qualitative conclusion is unchanged: the predictive power of beliefs increases with age even among respondents providing non-focal answers.

D.2. Differential Markups and Alternative Strategic Entry Mechanisms

A concern with interpreting these results as evidence of declining adverse selection is that insurers may charge differential markups across purchase ages that evolve over time. The identification strategy addresses this directly. By exploiting differences in how pricing evolves across age groups, common markup dynamics are absorbed. For the results to be driven by markups rather than selection, markup dynamics would need to differ systematically by purchase age in a way that correlates with private information. Markup compression is plausibly larger for younger, more elastic purchasers, thus the relative price reductions for older purchase ages work in a direction opposite to what entry patterns suggest. Figure 4B shows increasing entry over time, which should compress markups at all ages through competition; yet relative price reductions are largest for older purchasers. It is possible that markups do compress more for older

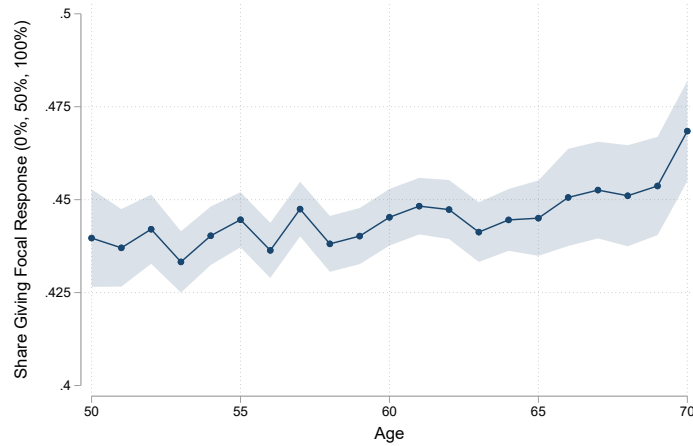


FIGURE A6. Focal Point Answers by Respondent Age

Data from the Health and Retirement Study, waves 1992–2018. Focal point answers defined as 0%, 50%, or 100%. Solid line reports sample fraction at each age pooling all cohorts and individuals. Shaded area reports 95% confidence interval.

	(1)	(2)
$\text{Belief}_{i,t}$	−0.012 (0.011)	0.004 (0.013)
$\text{Belief}_{i,t} \times \text{Age}_{i,t}$	−0.005*** (0.001)	−0.007*** (0.002)
Sample	<i>Full</i>	<i>Non – Focal</i>
N	109,316	58,147
R ²	0.204	0.203

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors clustered by individual. Dependent variable is an indicator for death within 10 years. Sample restricted to respondents aged 50–70 for whom mortality outcomes are resolved estimating equation A61. Data from the Health and Retirement Study, waves 1992–2018. Column 1 reproduces analysis from Table 4 in the main text. Column 2 excludes respondents providing focal point answers of 0%, 50%, or 100%. All columns additionally control for health, gender and race.

TABLE A11. Mortality-Relevant Private Information and Age

purchasers if they are less-elastic and face even higher initial markups than younger purchasers. However, there is no evidence to suggest this is the case. The time fixed effects in the main specification non-parametrically absorb all time-varying common factors, including any markup dynamics induced by firm entry; the differential estimate

is virtually unchanged when these are instead replaced with explicit aggregate-demand controls (Appendix C).

Two specific mechanisms could still generate differential dynamics. First, in a new market firms may be learning about the distribution of risks (i.e., cost types) and may do so differentially among the pools of potential annuitants at different purchase ages. This could lead to initial mispricing and induce a negative correlation. However, although deferred income annuities are a new product, insurers are experienced in pricing annuity and life-insurance contracts based on the distribution of underlying mortality risk. As equation (20) shows, the actuarial formula for a deferred income annuity payout is almost identical to the formulas for these existing products. They differ only in the timing at which payout begins. Therefore, it would have to be the case that insurers were learning about the declining adverse selection by age as the market matures which is exactly the fundamental prediction of the model.

Second, firms may initially ‘invest’ in market power, e.g. by undercutting or offering ‘sweetheart’ deals, and subsequently raise markups to ‘harvest’ downward sloping residual demand.¹² This would produce a negative contemporaneous correlation between equilibrium prices and demand, a distinct implication to the time path considered here. Additionally, this mechanism has sharp predictions about the declining number of insurers in the market which is at odds with Figure 4. The number of insurers offering these contracts grows over time—inconsistent with price and quantity dynamics driven by this form of strategic behavior. Moreover, such subsidization works against the finding. This form of strategic pricing raises money’s worth and so *lowers* measured selection precisely when the market is immature, making the observed pattern of stronger early selection conservative with respect to this channel.

¹²This may arise if consumers do not switch products due to search costs. This is unlikely to be of relevance here where the annuitization decision is typically once and for all.

References

- Chade, H. and E. Schlee (2012). Optimal insurance with adverse selection. *Theoretical Economics* 7(3), 571–607.
- Chade, H. and J. Swinkels (2021). Screening in vertical oligopolies. *Econometrica* 89(3), 1265–1311.
- Cherubini, U., S. Mulinacci, F. Gobbi, and S. Romagnoli (2011). *Dynamic Copula Methods in Finance*. John Wiley & Sons.
- Elder, T. E. (2013). The predictive validity of subjective mortality expectations: Evidence from the health and retirement study. *Demography* 50(2), 569–589.
- Fang, Z. and H. Joe (1992). Further developments on some dependence orderings for continuous bivariate distributions. *Annals of the Institute of Statistical Mathematics* 44, 501–517.
- Foltyn, R. and J. Olsson (2024). Subjective life expectancies, time preference heterogeneity, and wealth inequality. *Quantitative Economics* 15(3), 699–736.
- Gürkaynak, R. S., B. Sack, and J. H. Wright (2007). The us treasury yield curve: 1961 to the present. *Journal of monetary Economics* 54(8), 2291–2304.
- Hamermesh, D. S. (1985). Expectations, life expectancy, and economic behavior. *The Quarterly Journal of Economics* 100(2), 389–408.
- Hurd, M. D. and K. McGarry (2002). The predictive validity of subjective probabilities of survival. *The Economic Journal* 112(482), 966–985.
- Levy, H. (1982). Stochastic dominance rules for truncated normal distributions: A note. *The Journal of Finance* 37(5), 1299–1303.
- MacKinnon, J. G., M. Ø. Nielsen, and M. D. Webb (2021). Wild bootstrap and asymptotic inference with multiway clustering. *Journal of Business & Economic Statistics* 39(2), 505–519.
- Milgrom, P. R. (1981). Good news and bad news: Representation theorems and applications. *The Bell Journal of Economics*, 380–391.
- Milgrom, P. R. and R. J. Weber (1982). A theory of auctions and competitive bidding. *Econometrica*, 1089–1122.
- Mitchell, O. S., J. M. Poterba, M. J. Warshawsky, and J. R. Brown (1999). New evidence on the money's worth of individual annuities. *American Economic Review* 89(5), 1299–1318.
- Navarro, J. and J. M. Sarabia (2022). Copula representations for the sum of dependent risks: models and comparisons. *Probability in the Engineering and Informational Sciences* 36(2), 320–340.
- Nelsen, R. B. (2006). *An Introduction to Copulas*. Springer.
- Poterba, J. M. and A. Solomon (2021). Discount rates, mortality projections, and money's worth calculations for us individual annuities. *NBER Working Paper*.
- Shaked, M. and J. G. Shanthikumar (2007). *Stochastic orders*. Springer.
- Singh, S. K. and G. S. Maddala (1976). A function for size distribution of incomes. *Econometrica* 44(5), 963–970.